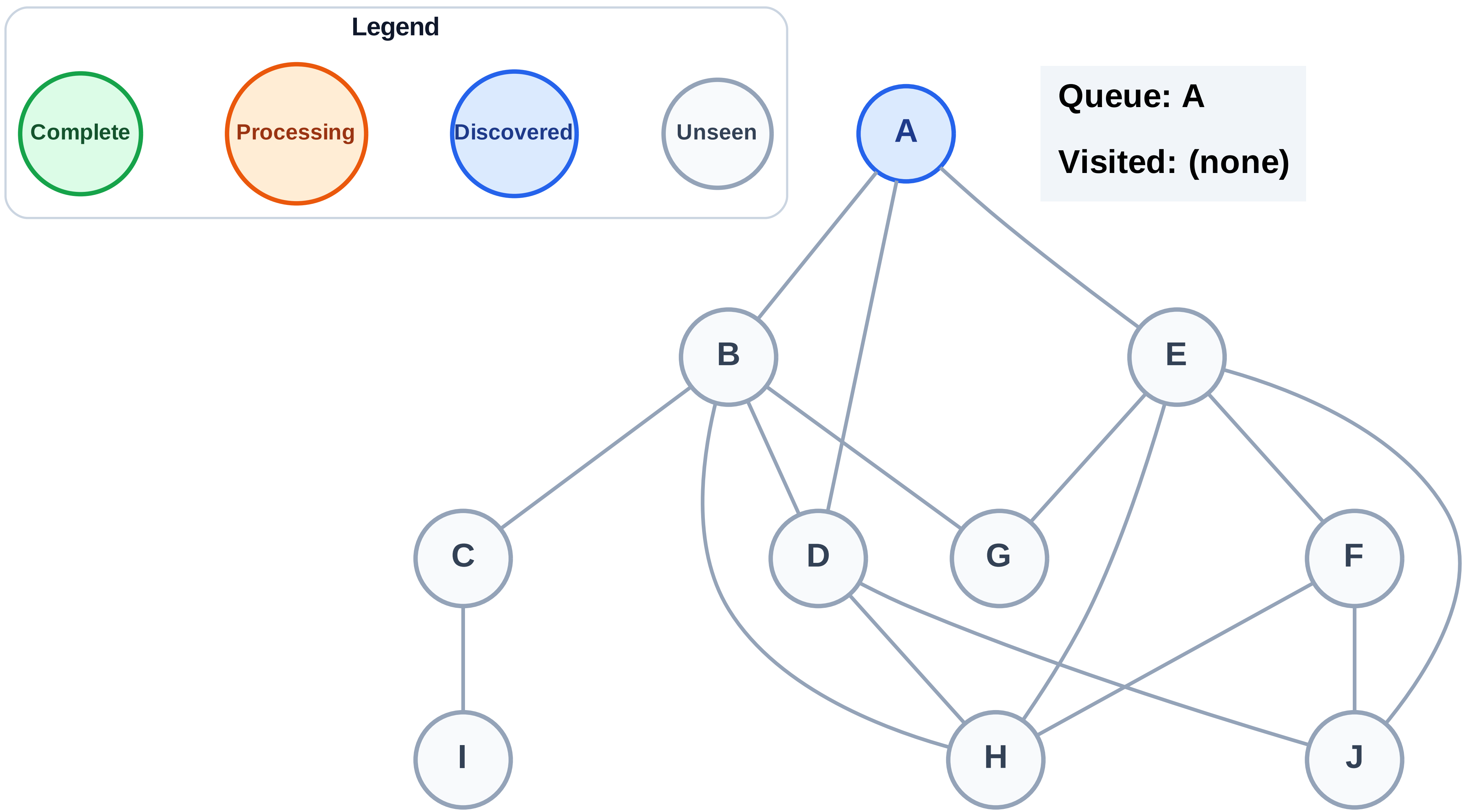
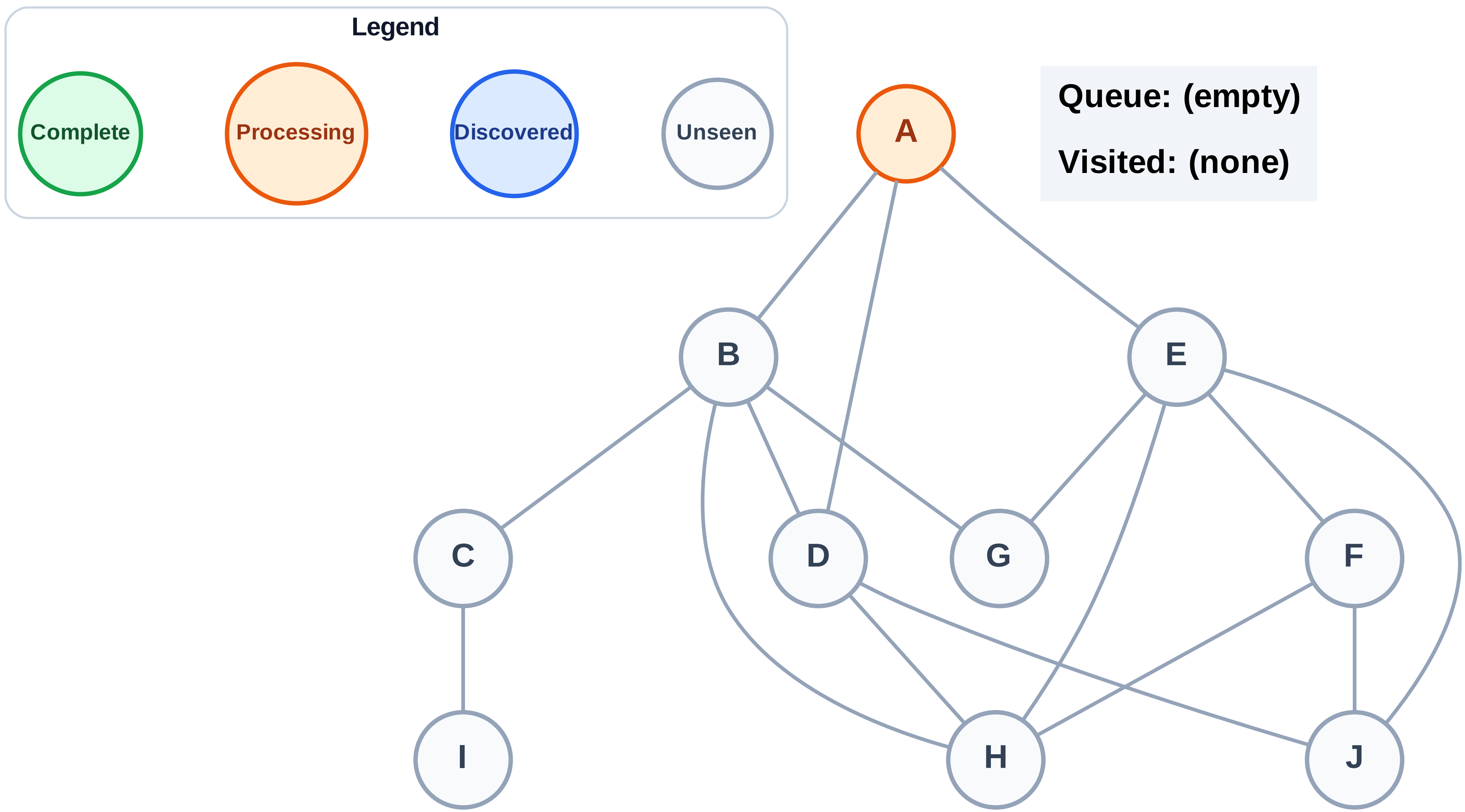
BFS Traversal

Step 1: Start at A



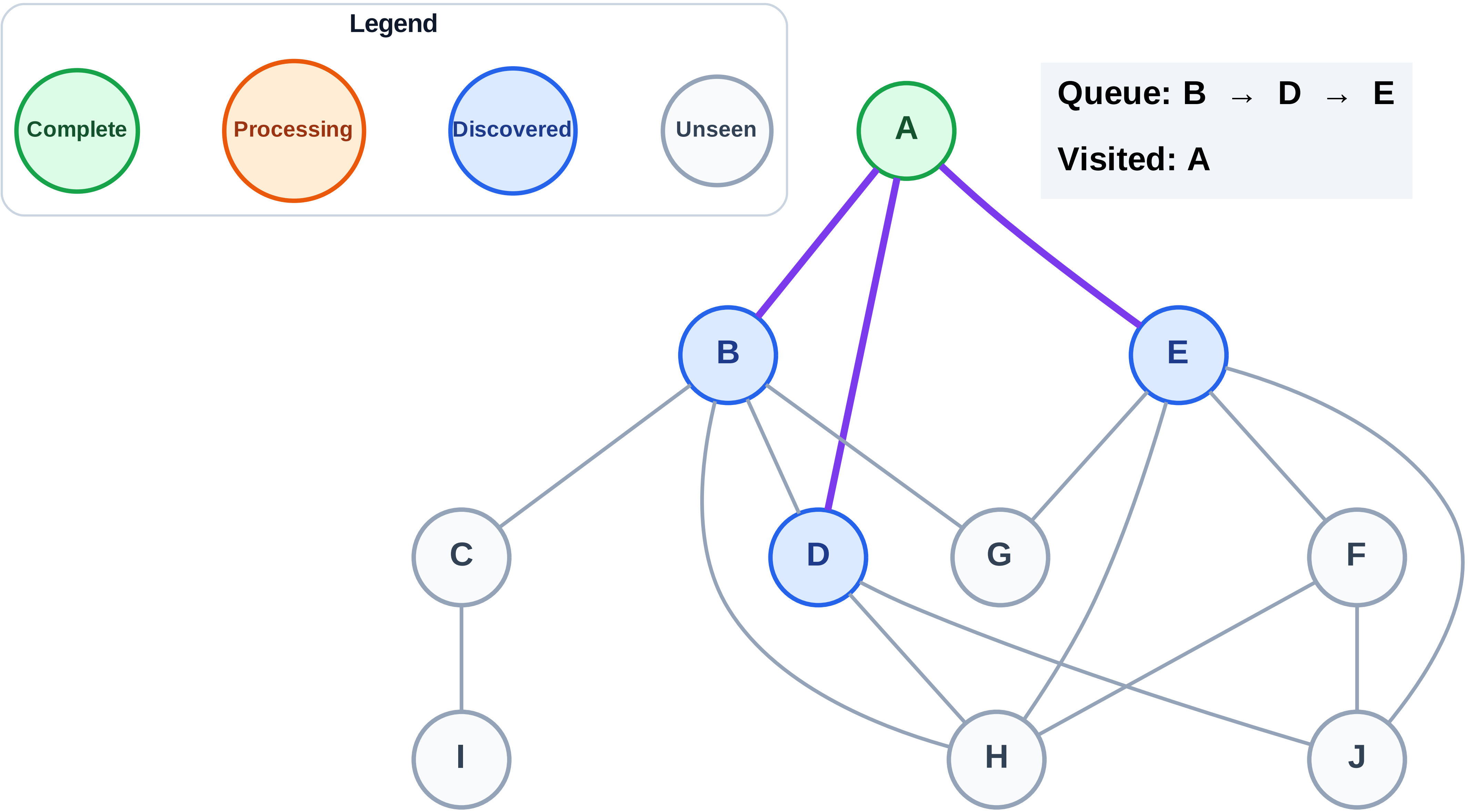
BFS Traversal

Step 2: Process node A



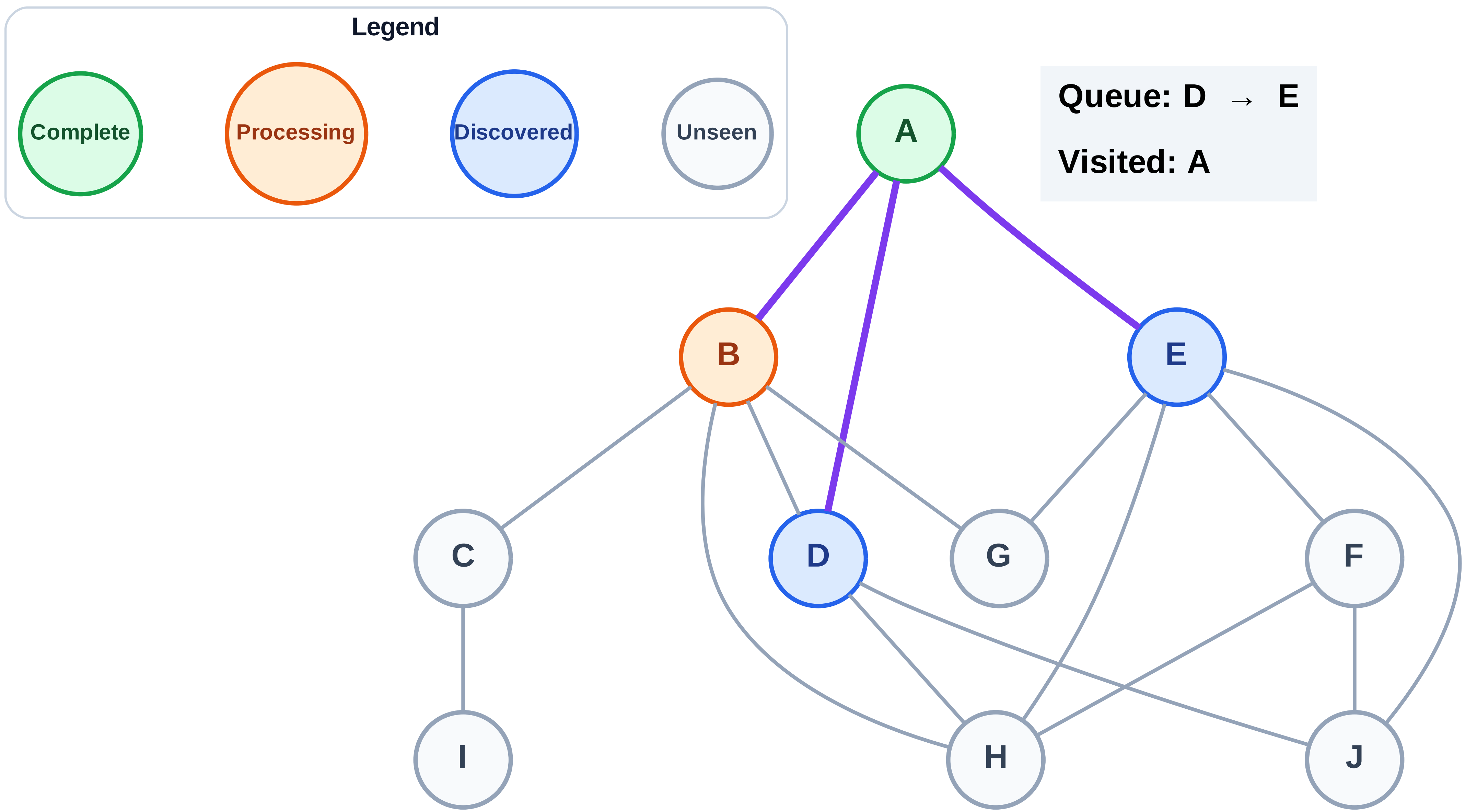
BFS Traversal

Step 3: Discover B, D, E from A



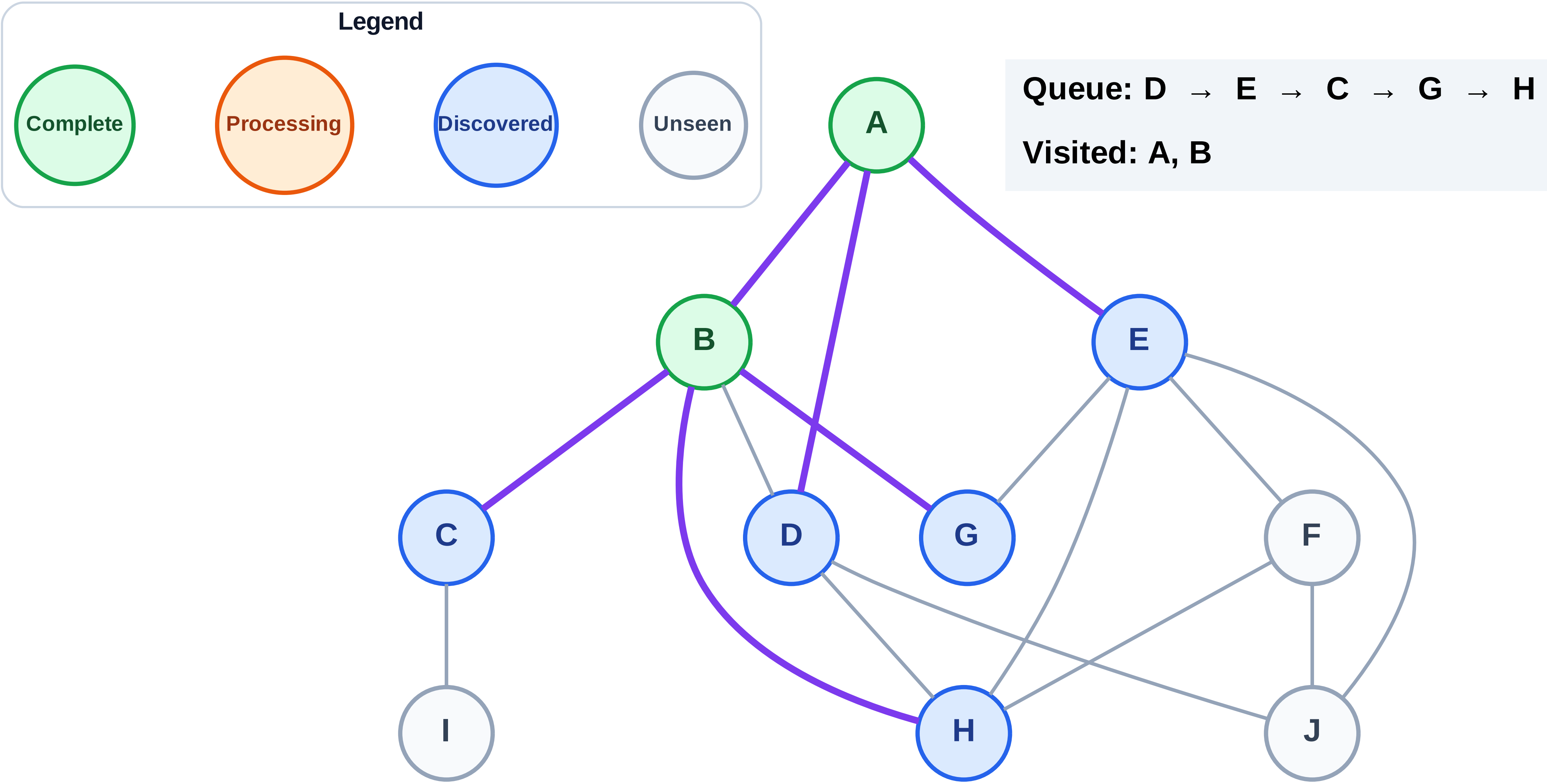
BFS Traversal

Step 4: Process node B



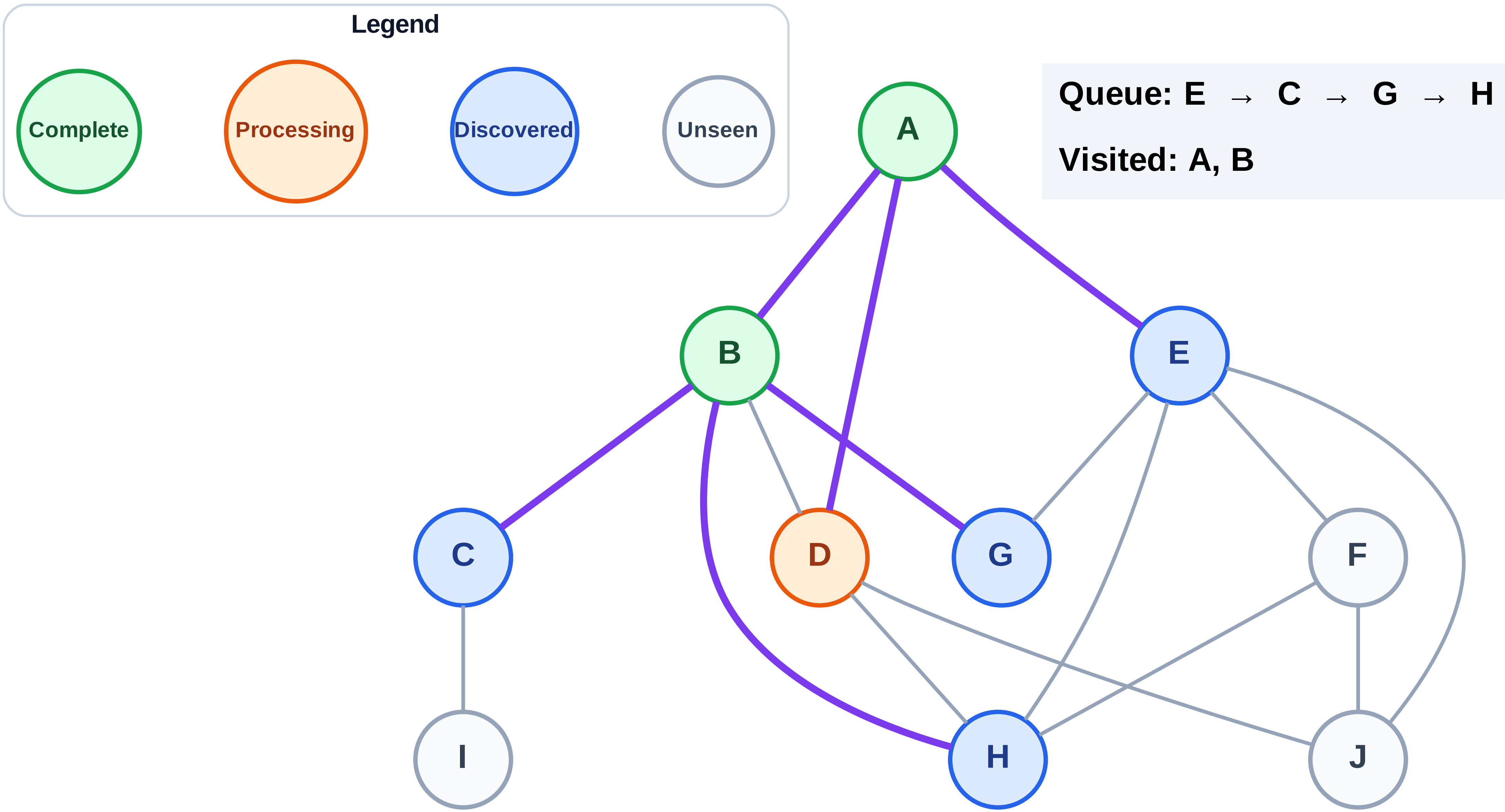
BFS Traversal

Step 5: Discover C, G, H from B



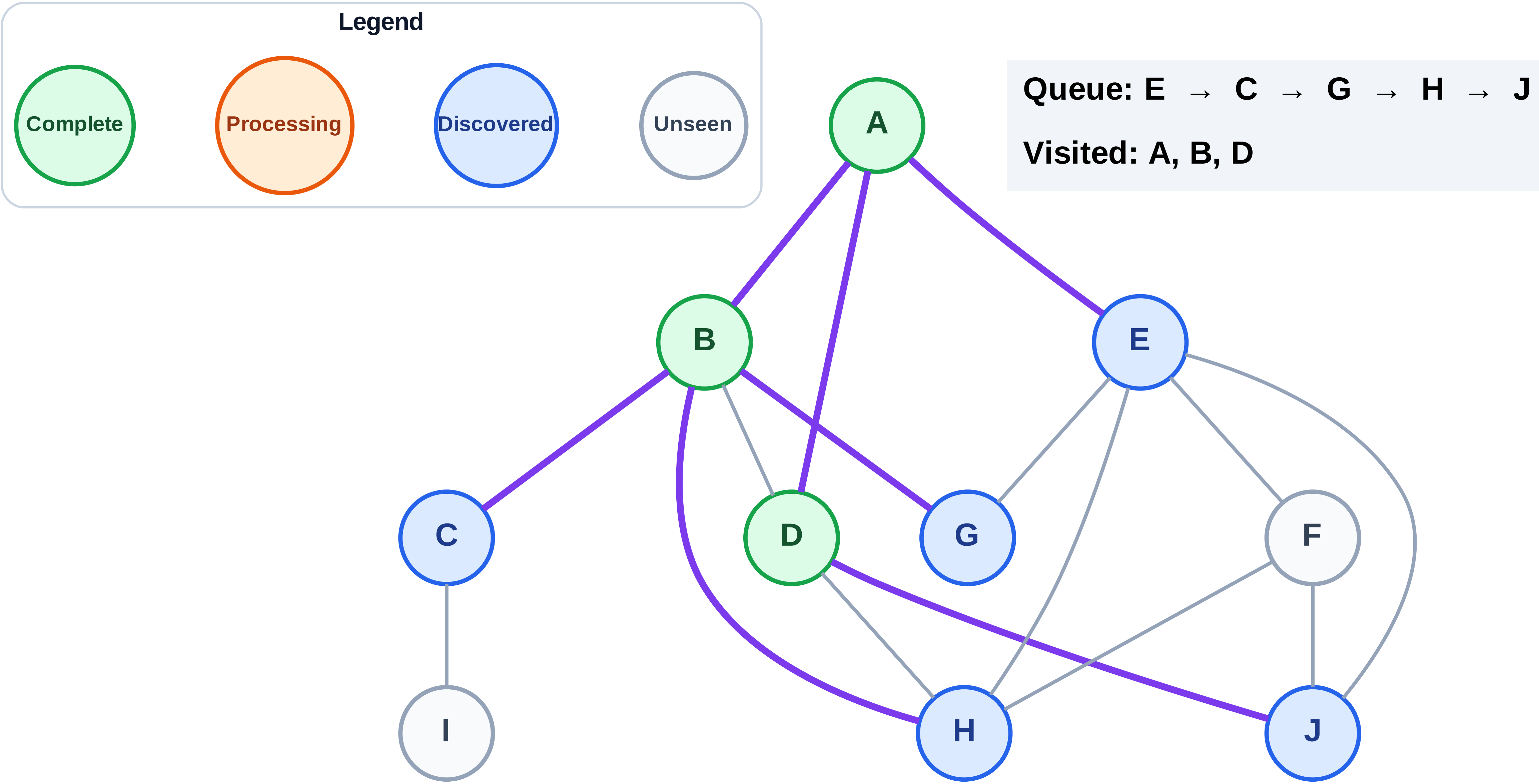
BFS Traversal

Step 6: Process node D



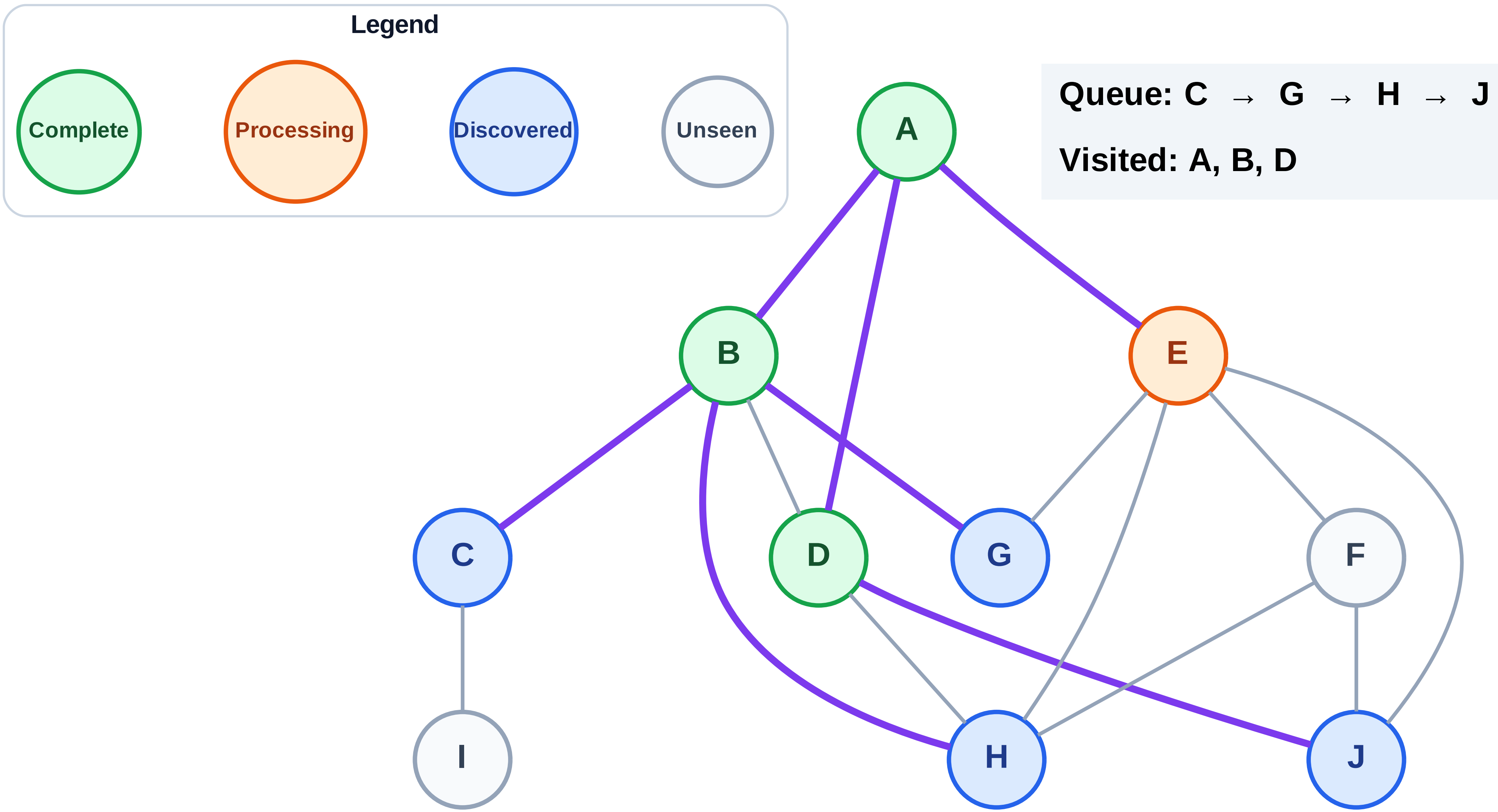
BFS Traversal

Step 7: Discover J from D



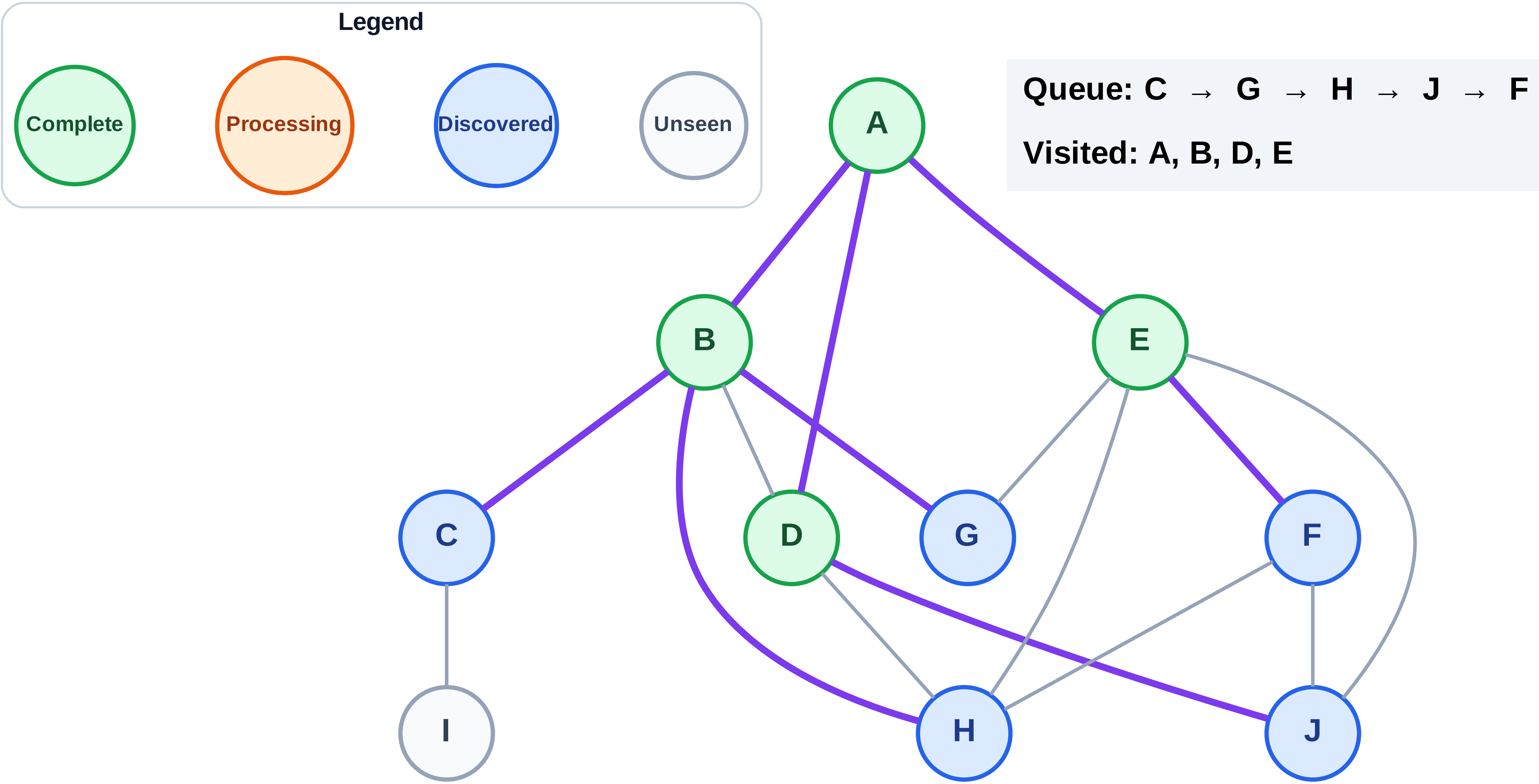
BFS Traversal

Step 8: Process node E



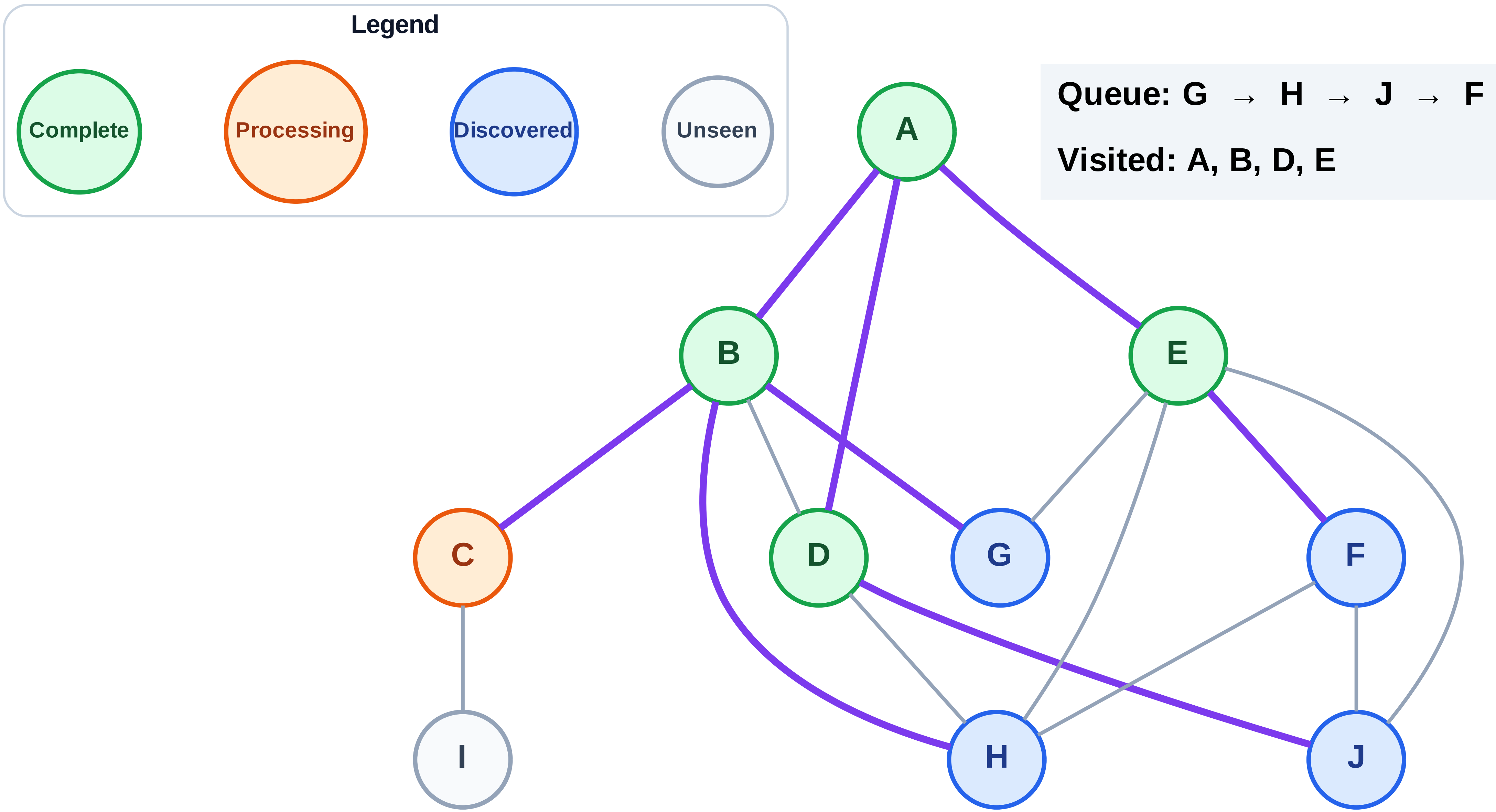
BFS Traversal

Step 9: Discover F from E



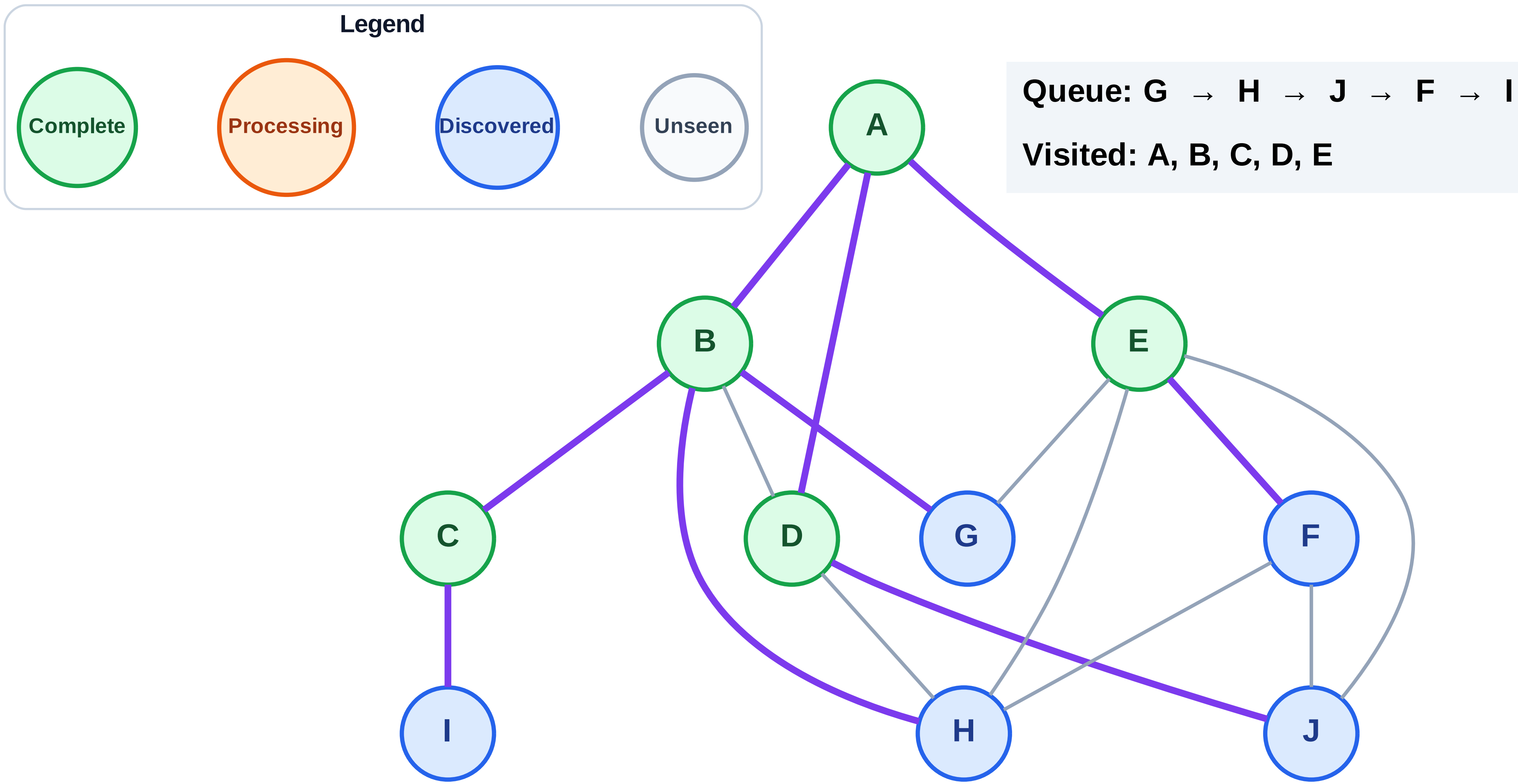
BFS Traversal

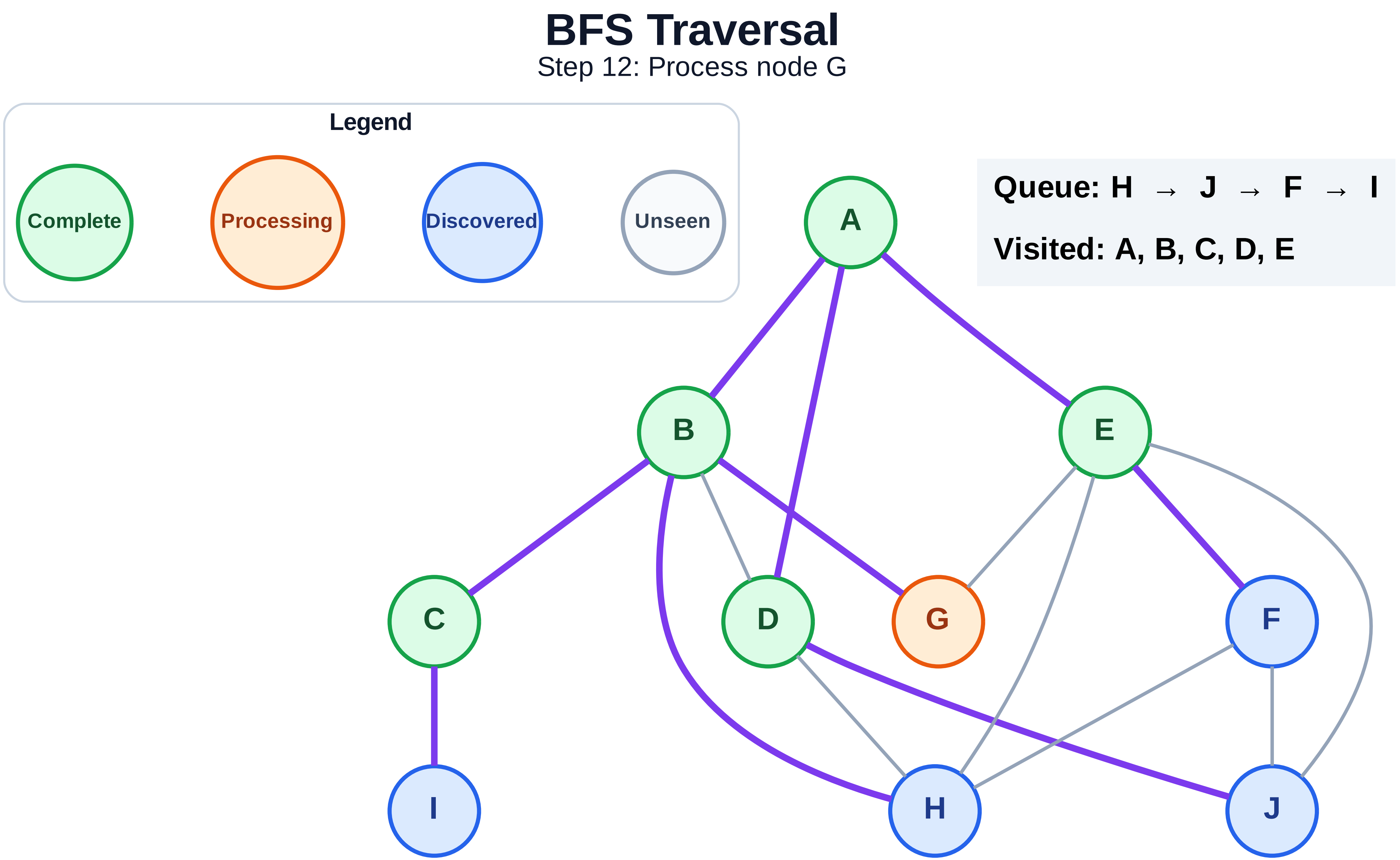
Step 10: Process node C

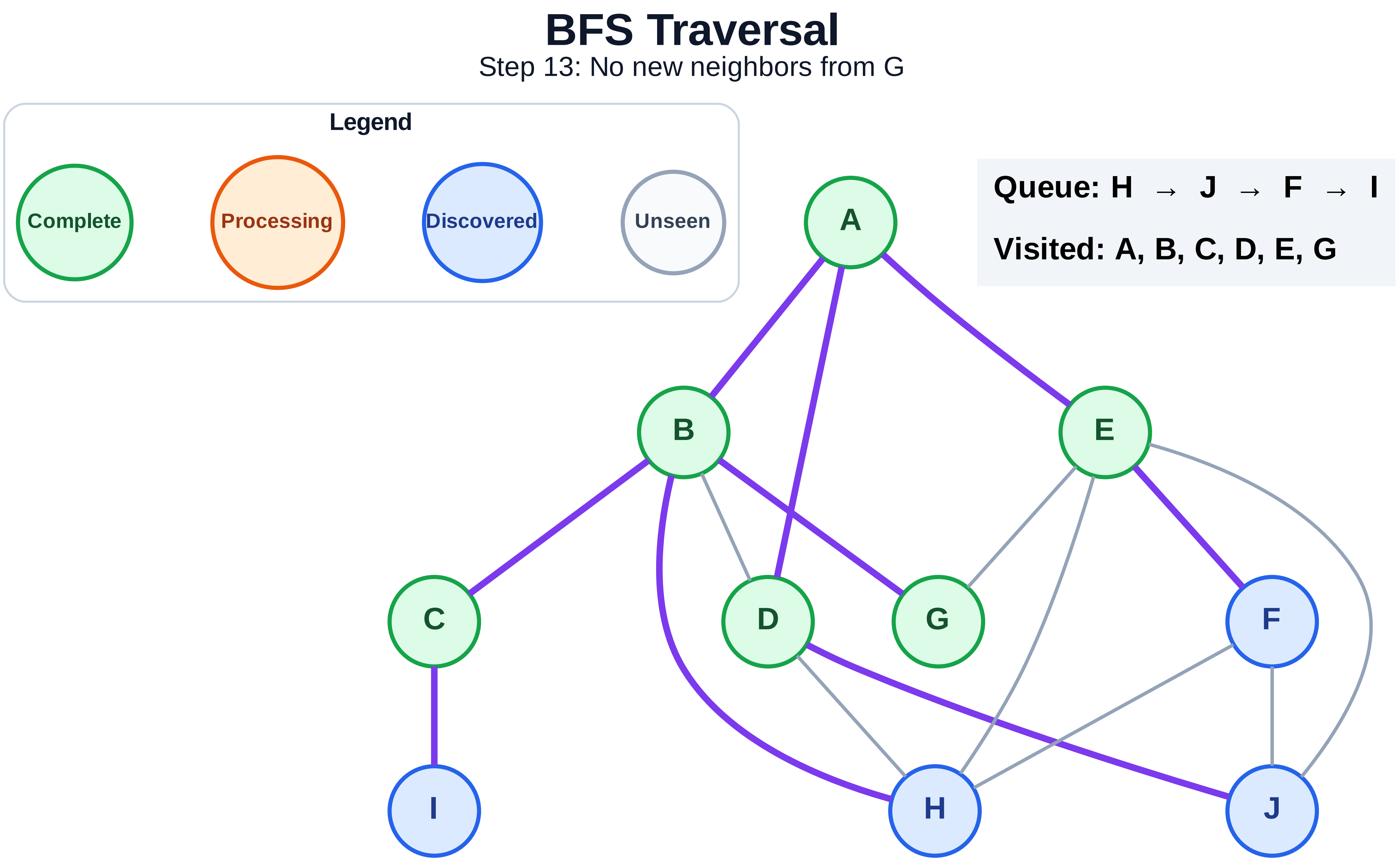


BFS Traversal

Step 11: Discover I from C

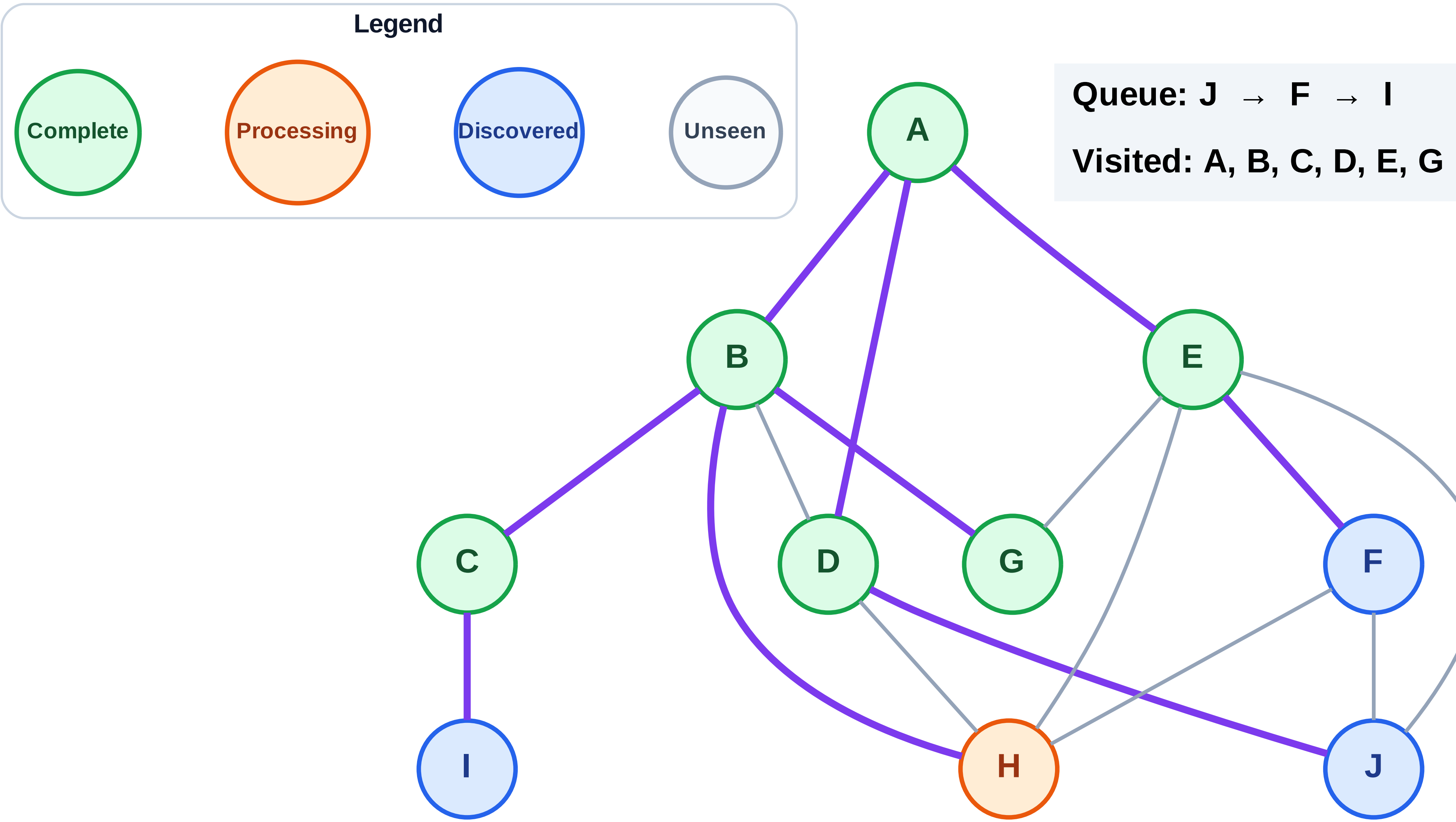



BFS Traversal
Step 12: Process node G
Legend
Complete
Processing
Discovered
Unseen
Queue: H → J → F → I
Visited: A, B, C, D, E
A
B
E
C
D
G
F
I
H
J


BFS Traversal
Step 13: No new neighbors from G
Legend
Complete
Processing
Discovered
Unseen
Queue: H → J → F → I
Visited: A, B, C, D, E, G
A
B
E
C
D
G
F
I
H
J

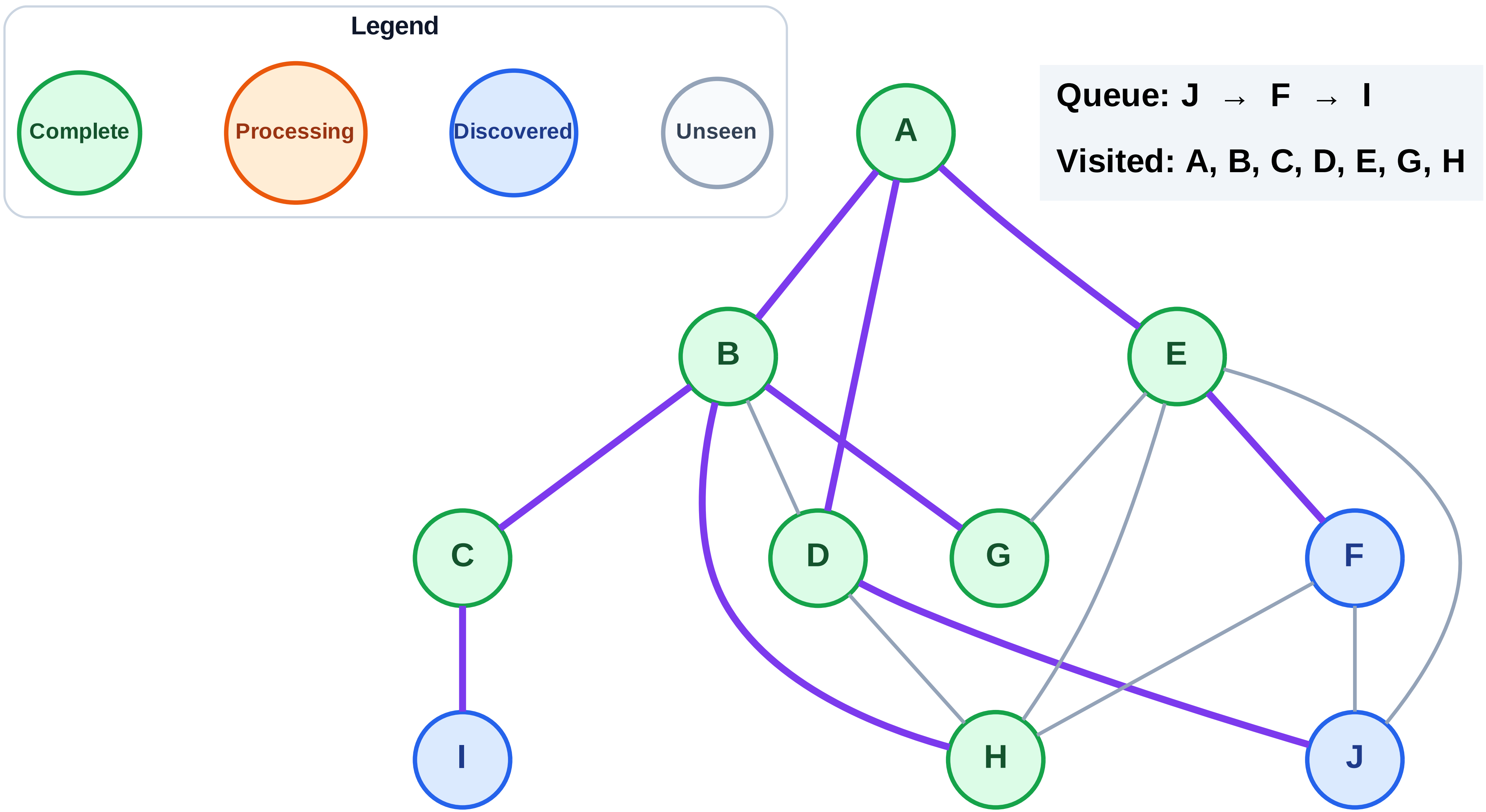
BFS Traversal

Step 14: Process node H



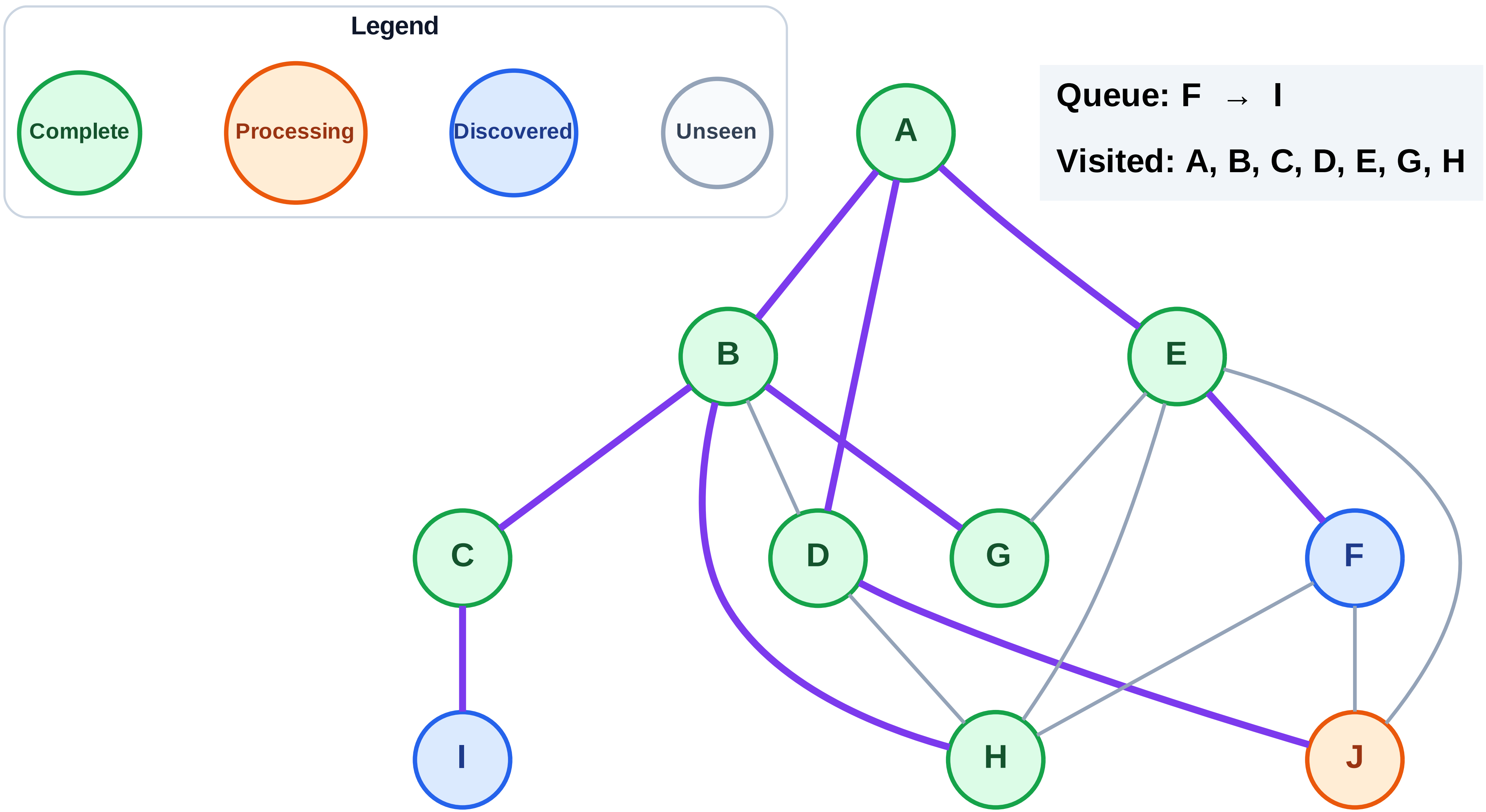
BFS Traversal

Step 15: No new neighbors from H



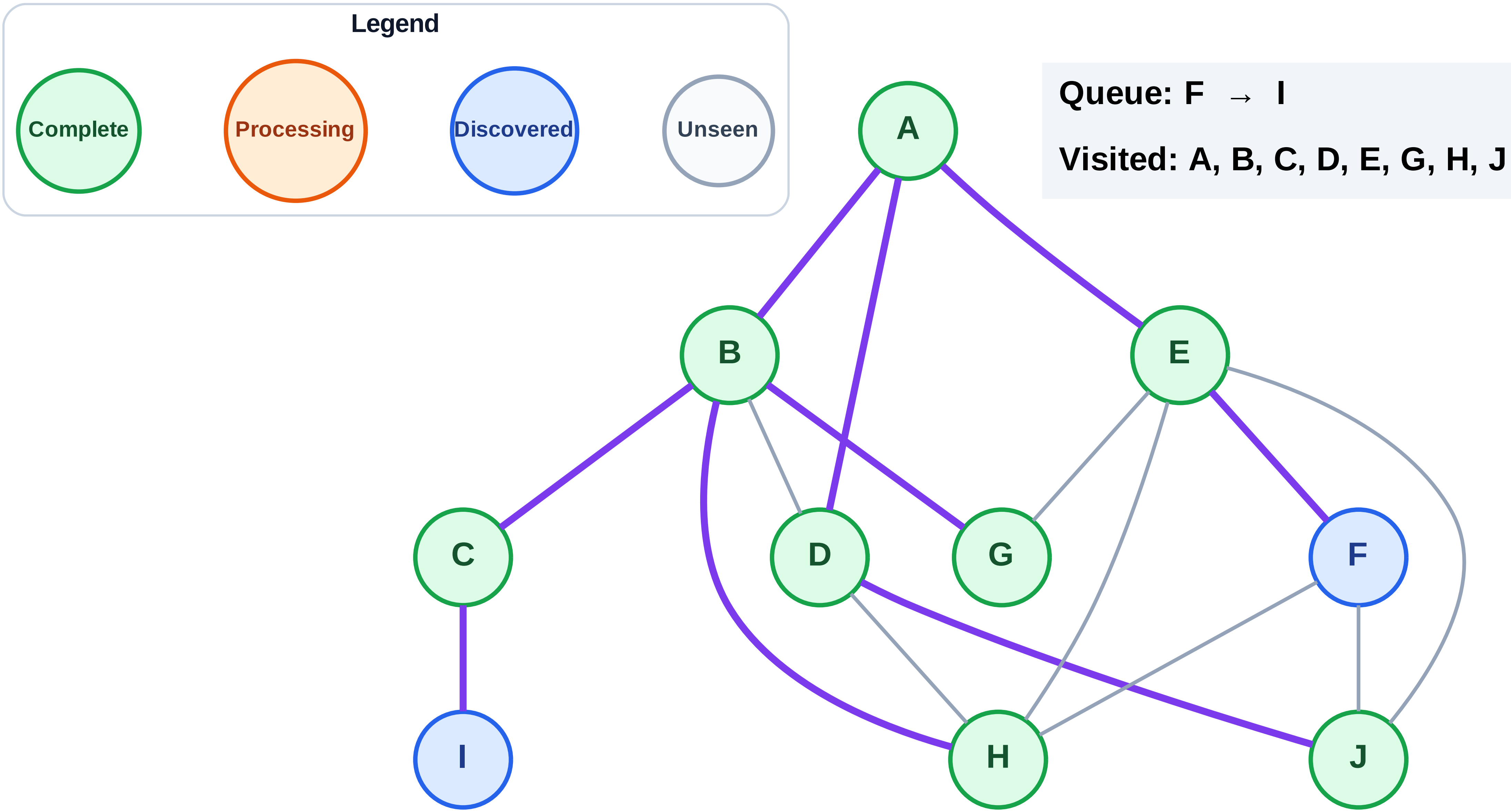
BFS Traversal

Step 16: Process node J



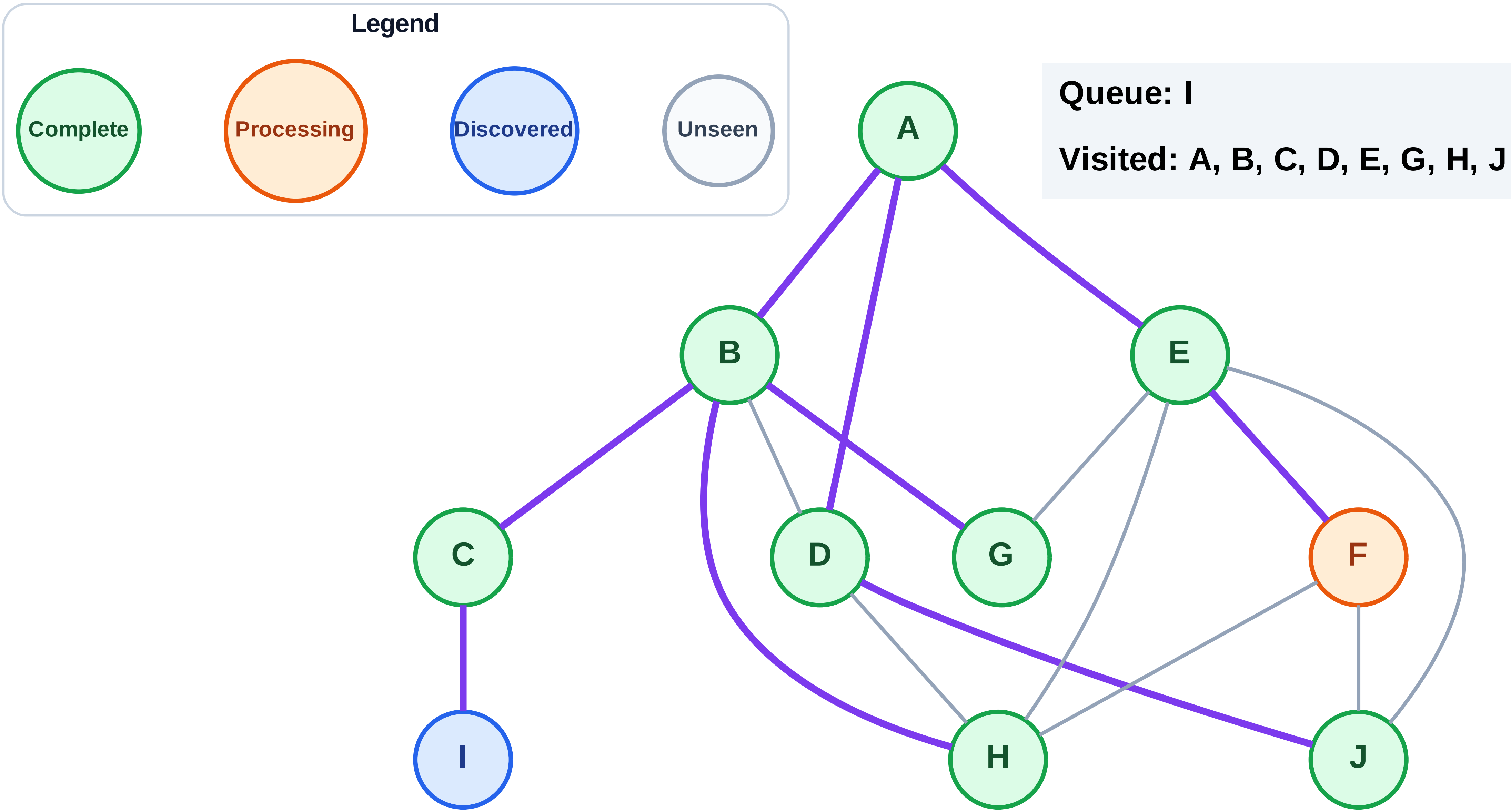
BFS Traversal

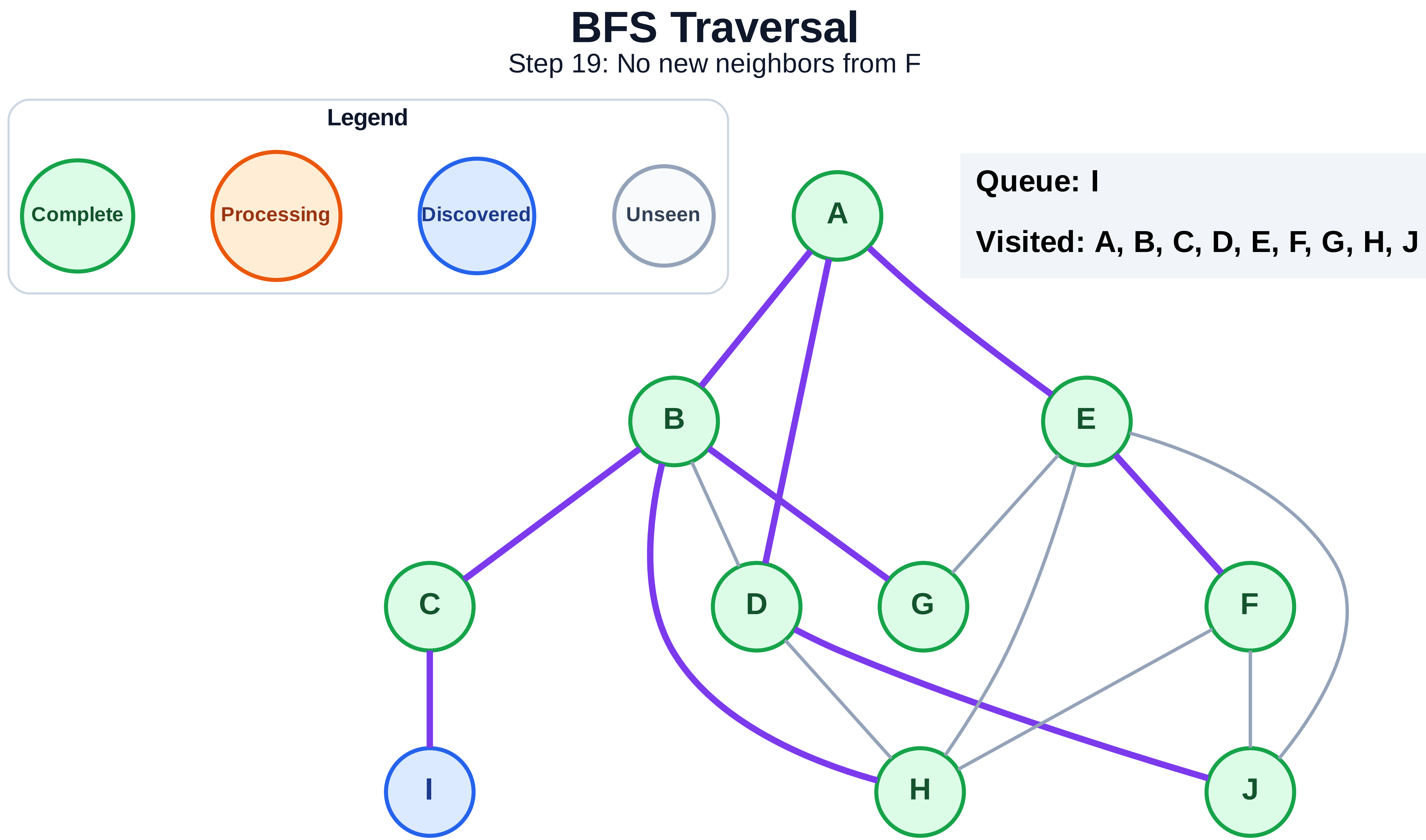
Step 17: No new neighbors from J



BFS Traversal

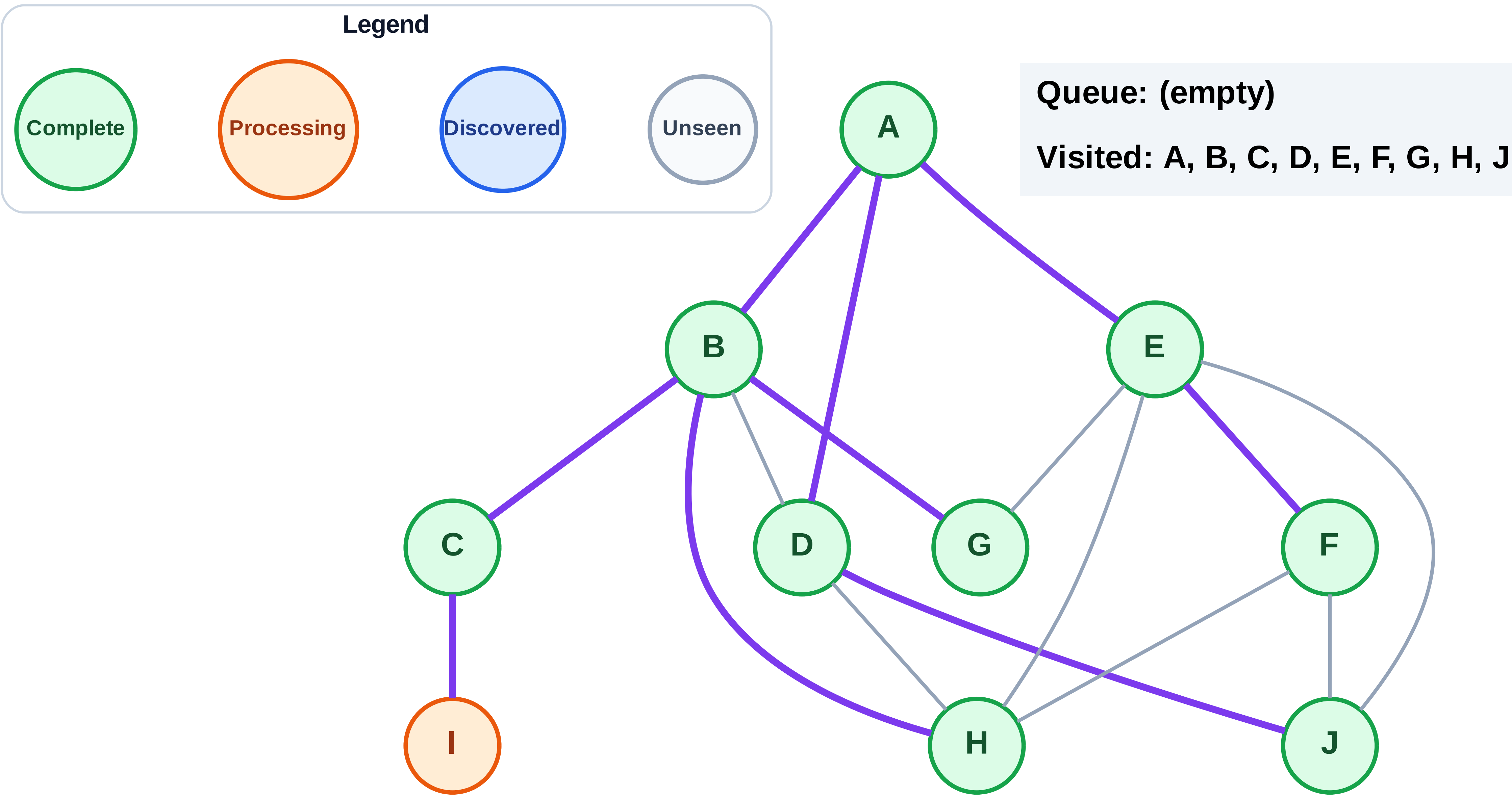
Step 18: Process node F




BFS Traversal
Step 19: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Queue: I
Visited: A, B, C, D, E, F, G, H, J
A
B
E
C
D
G
F
I
H
J

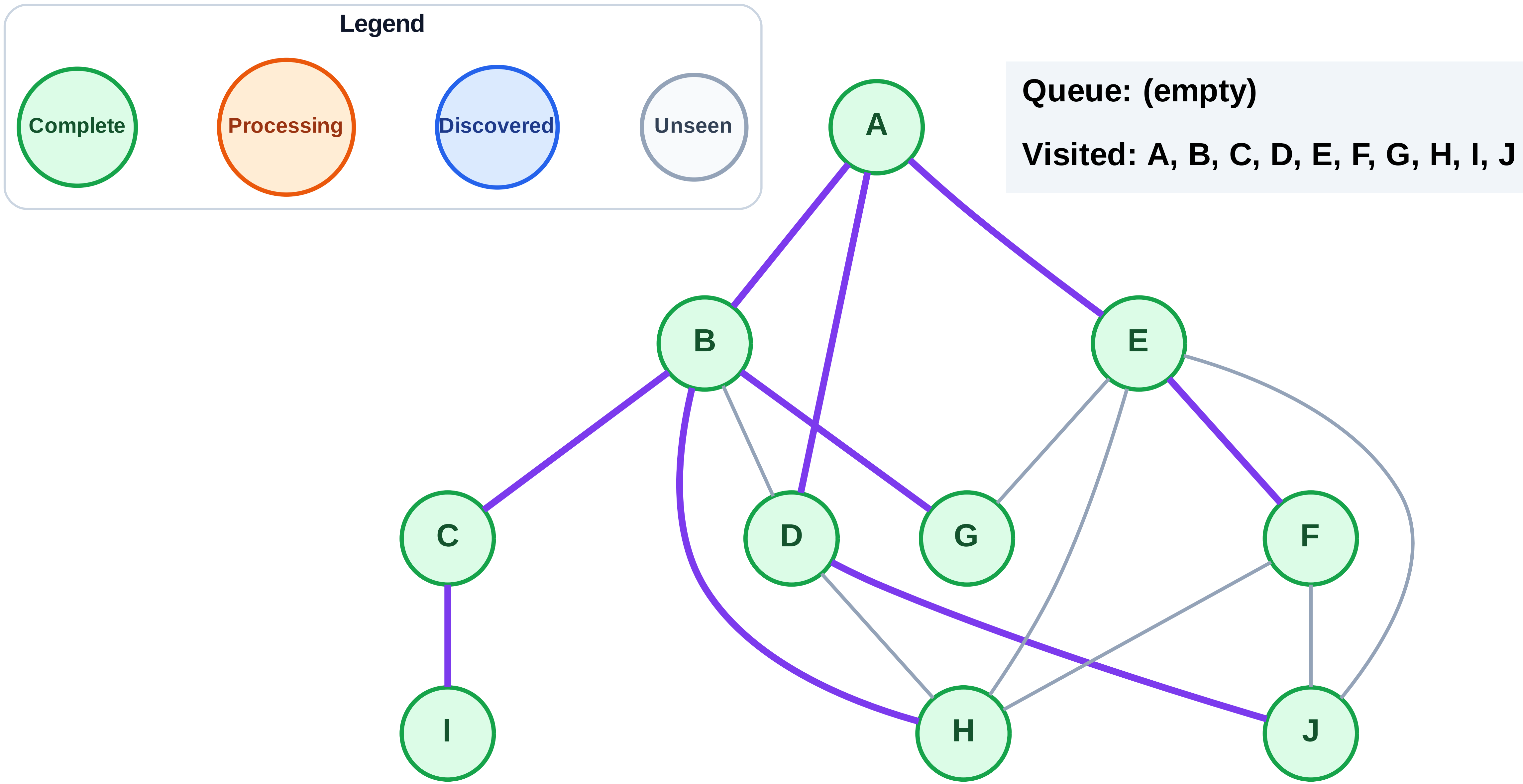
BFS Traversal

Step 20: Process node I



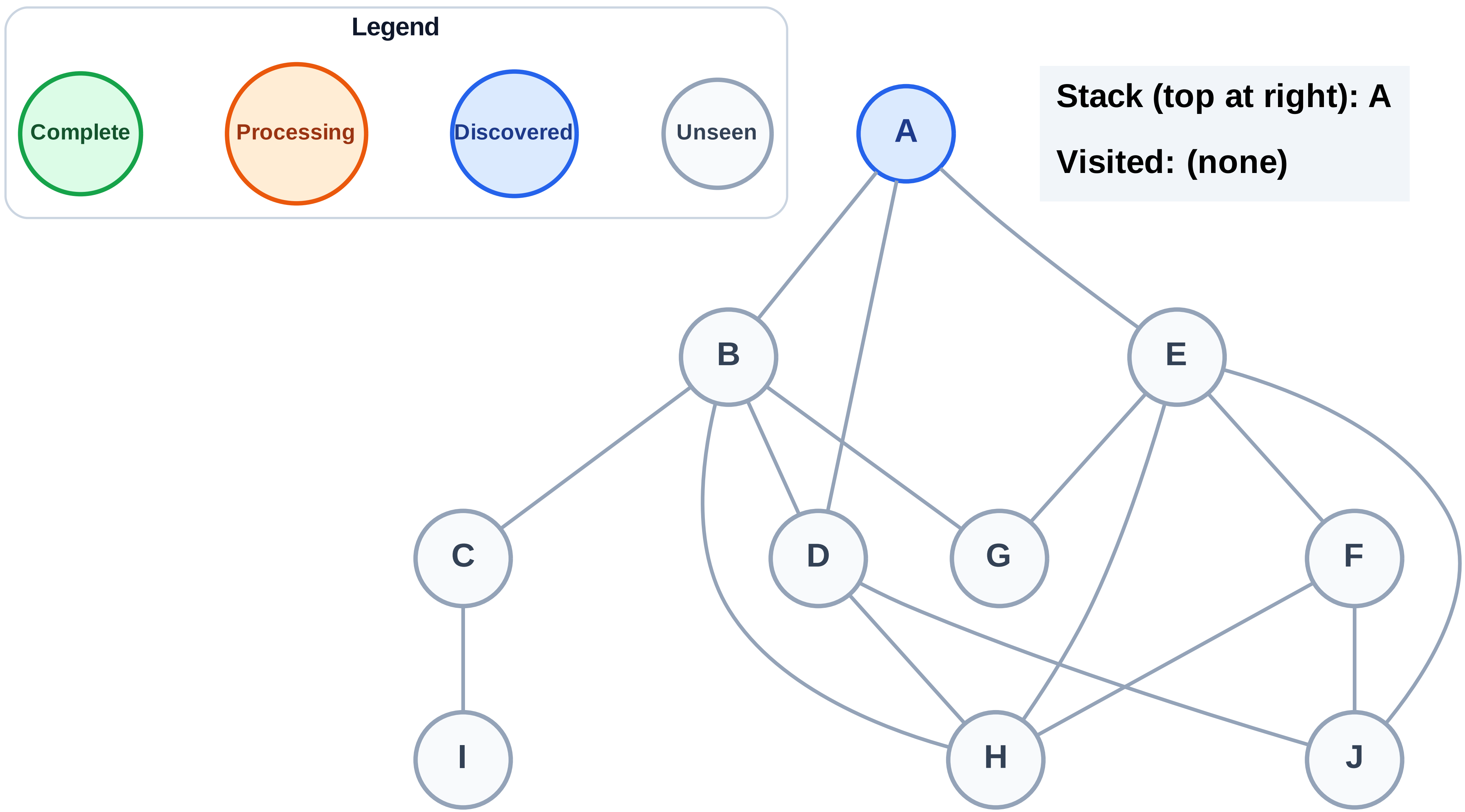
BFS Traversal

Step 21: No new neighbors from I



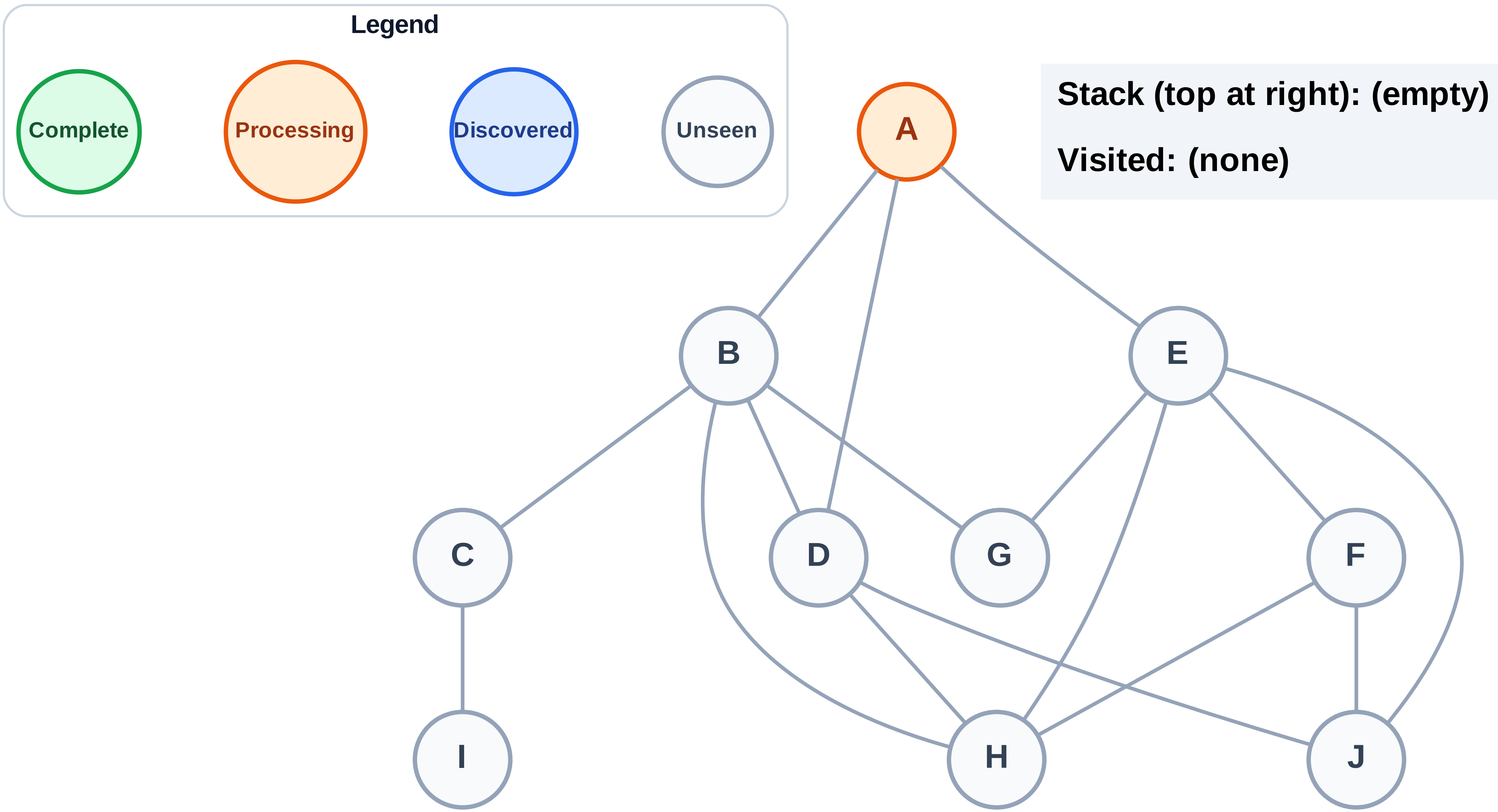
DFS Traversal

Step 1: Start at A



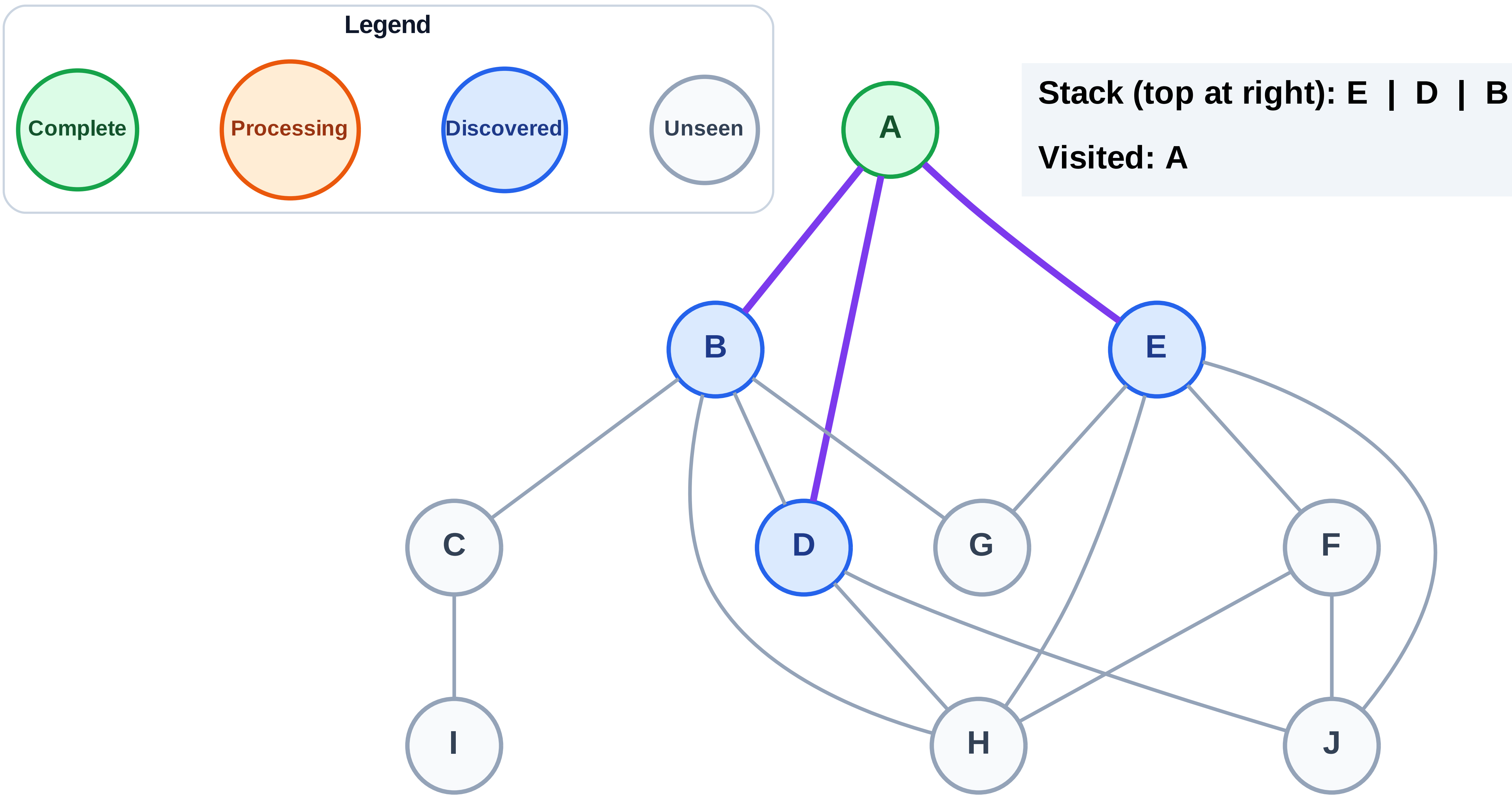
DFS Traversal

Step 2: Process node A



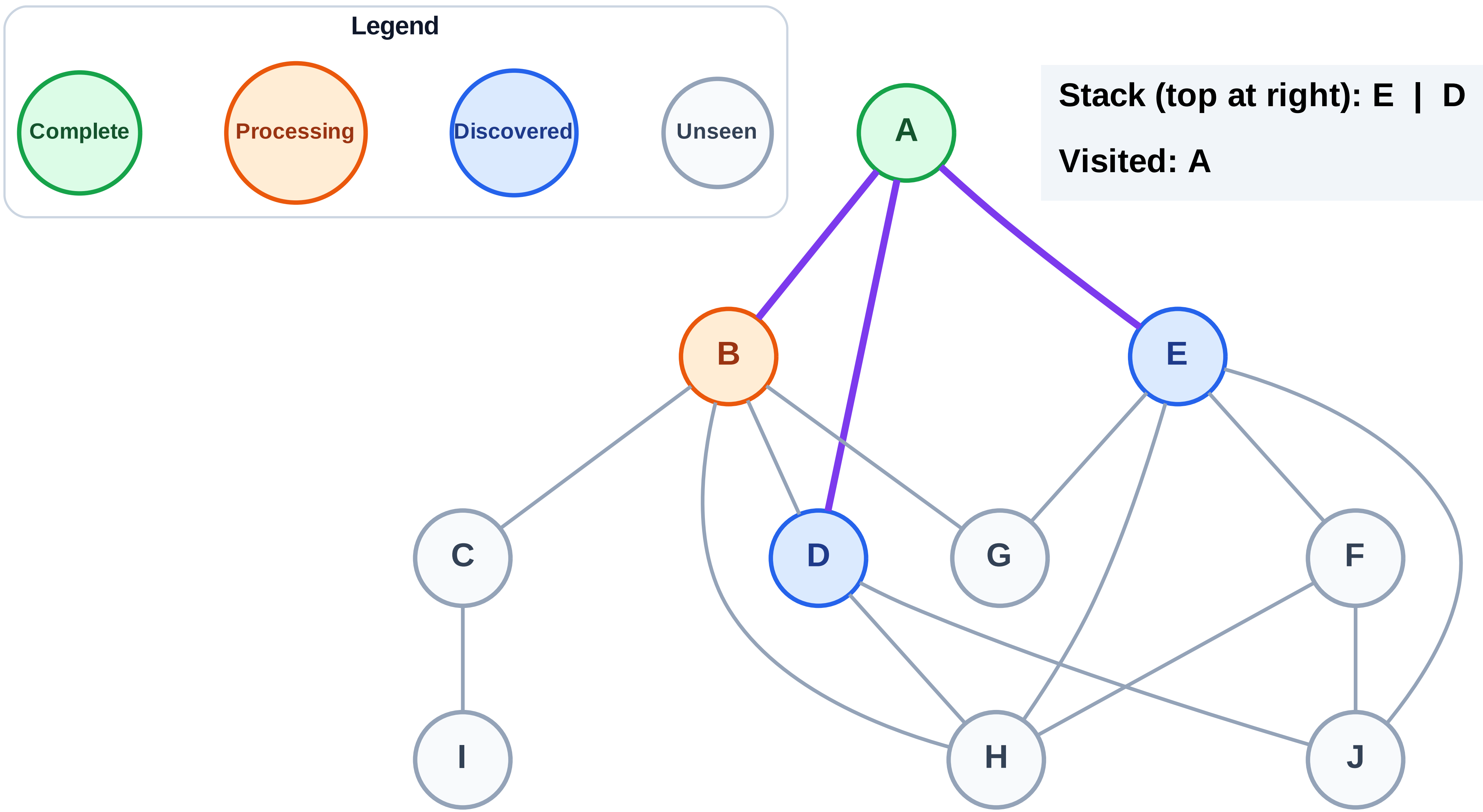
DFS Traversal

Step 3: Discover E, D, B from A



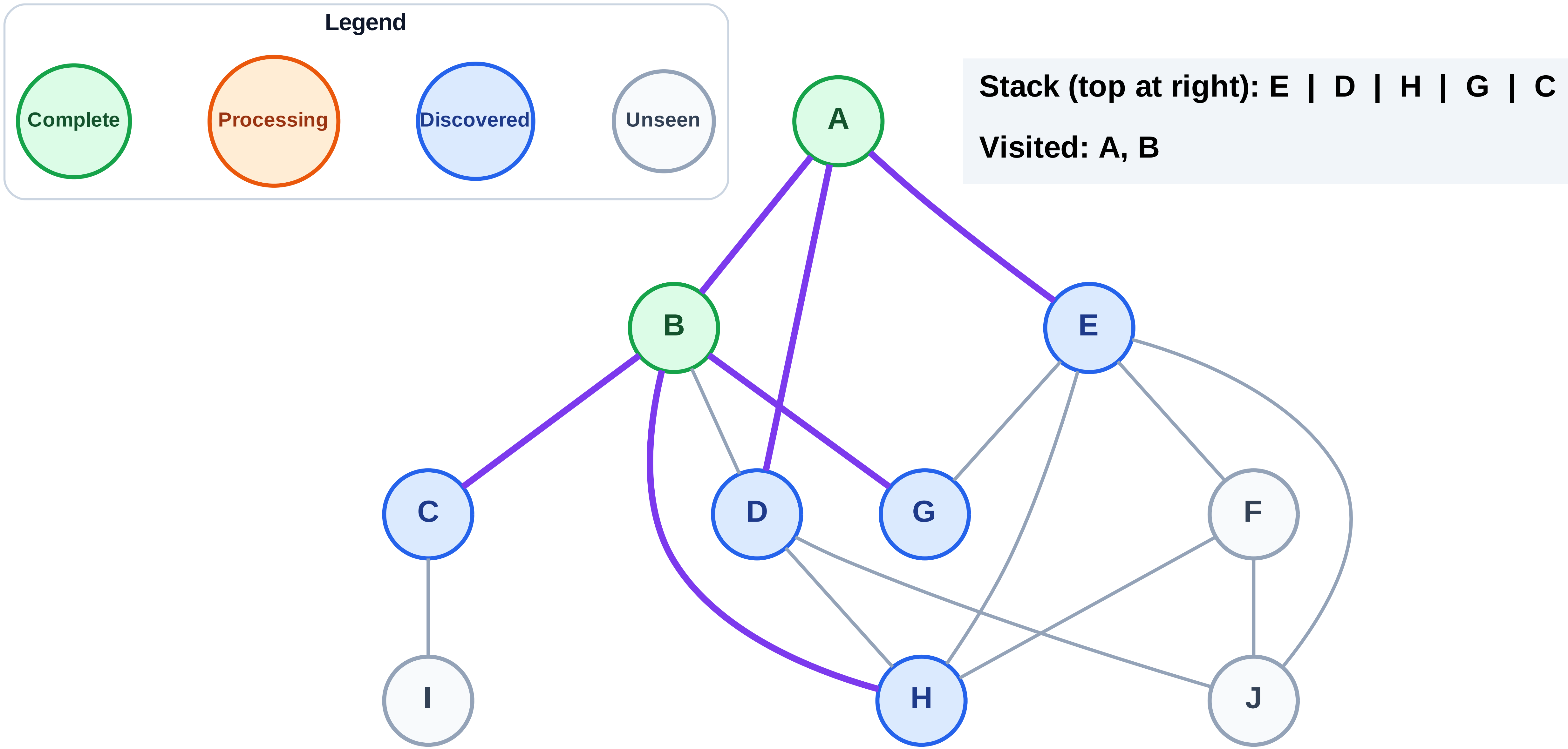
DFS Traversal

Step 4: Process node B



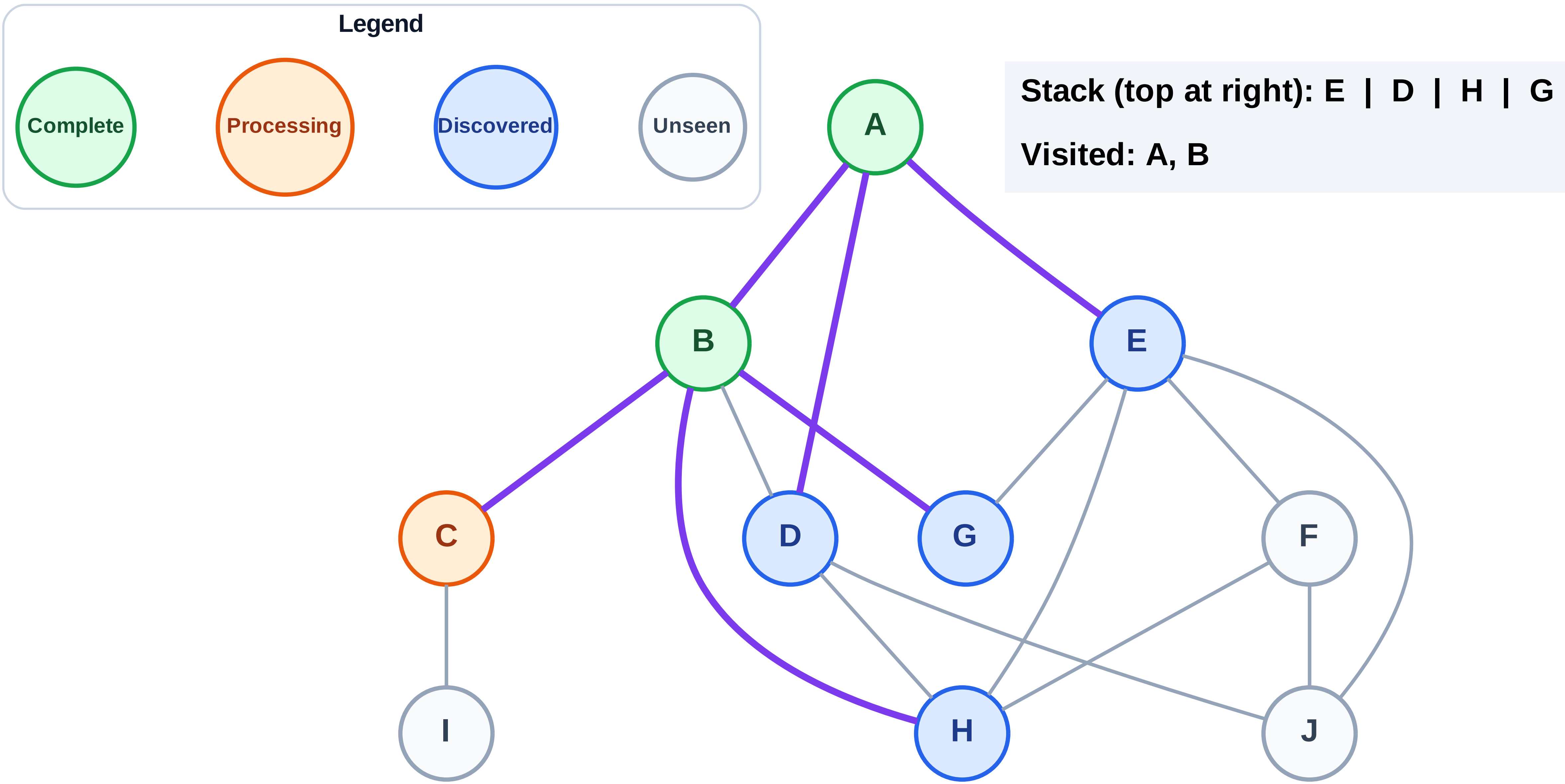
DFS Traversal

Step 5: Discover H, G, C from B



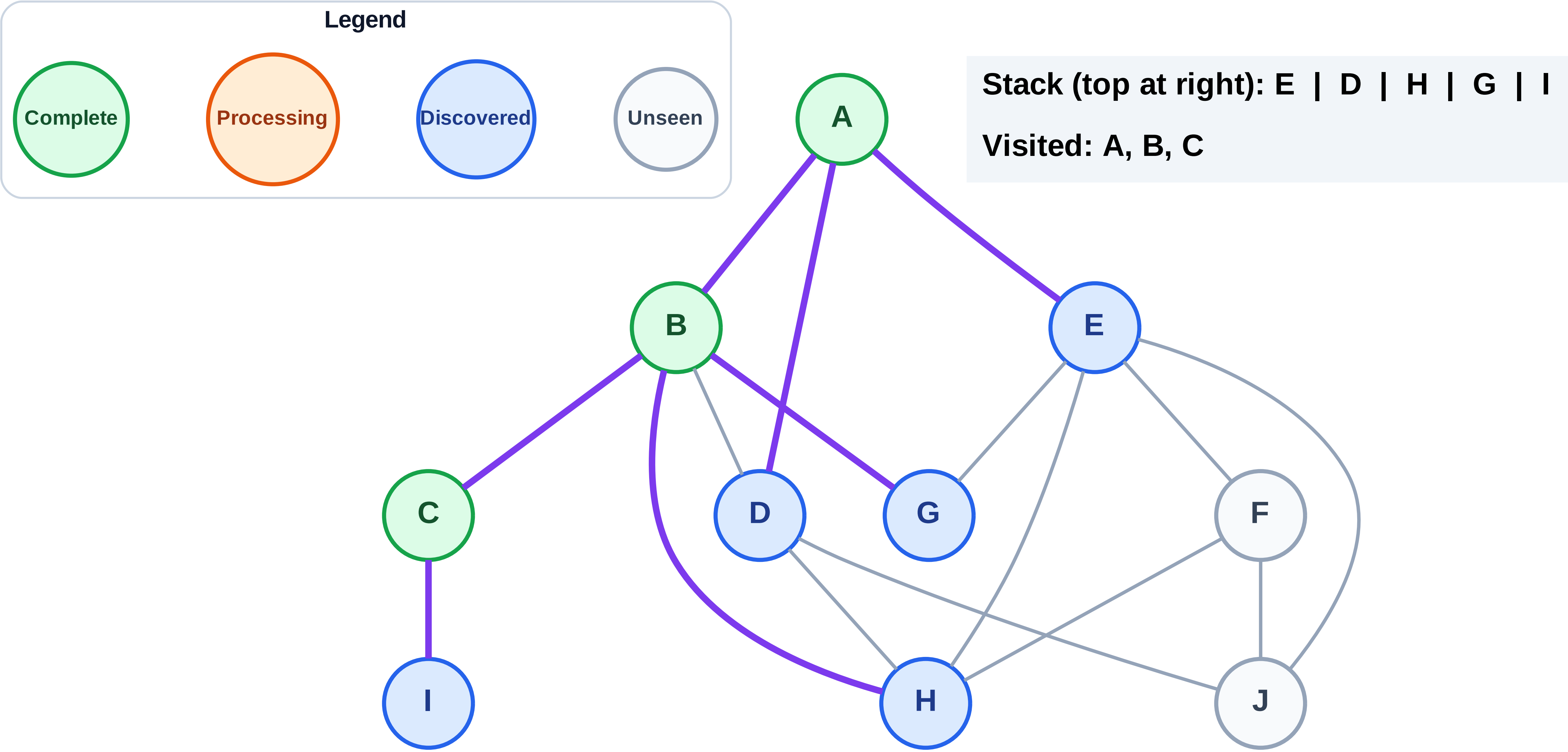
DFS Traversal

Step 6: Process node C



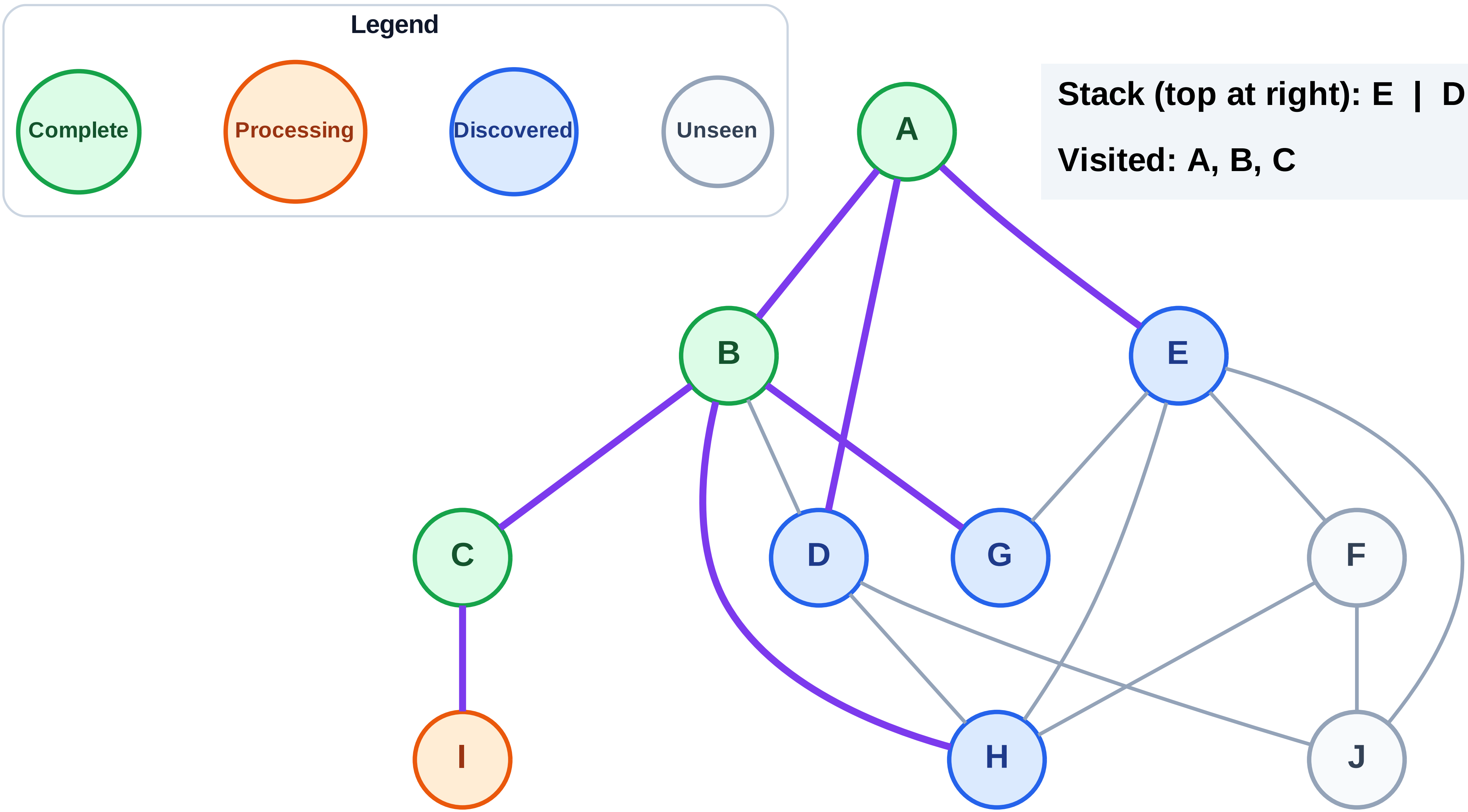
DFS Traversal

Step 7: Discover I from C



DFS Traversal

Step 8: Process node I

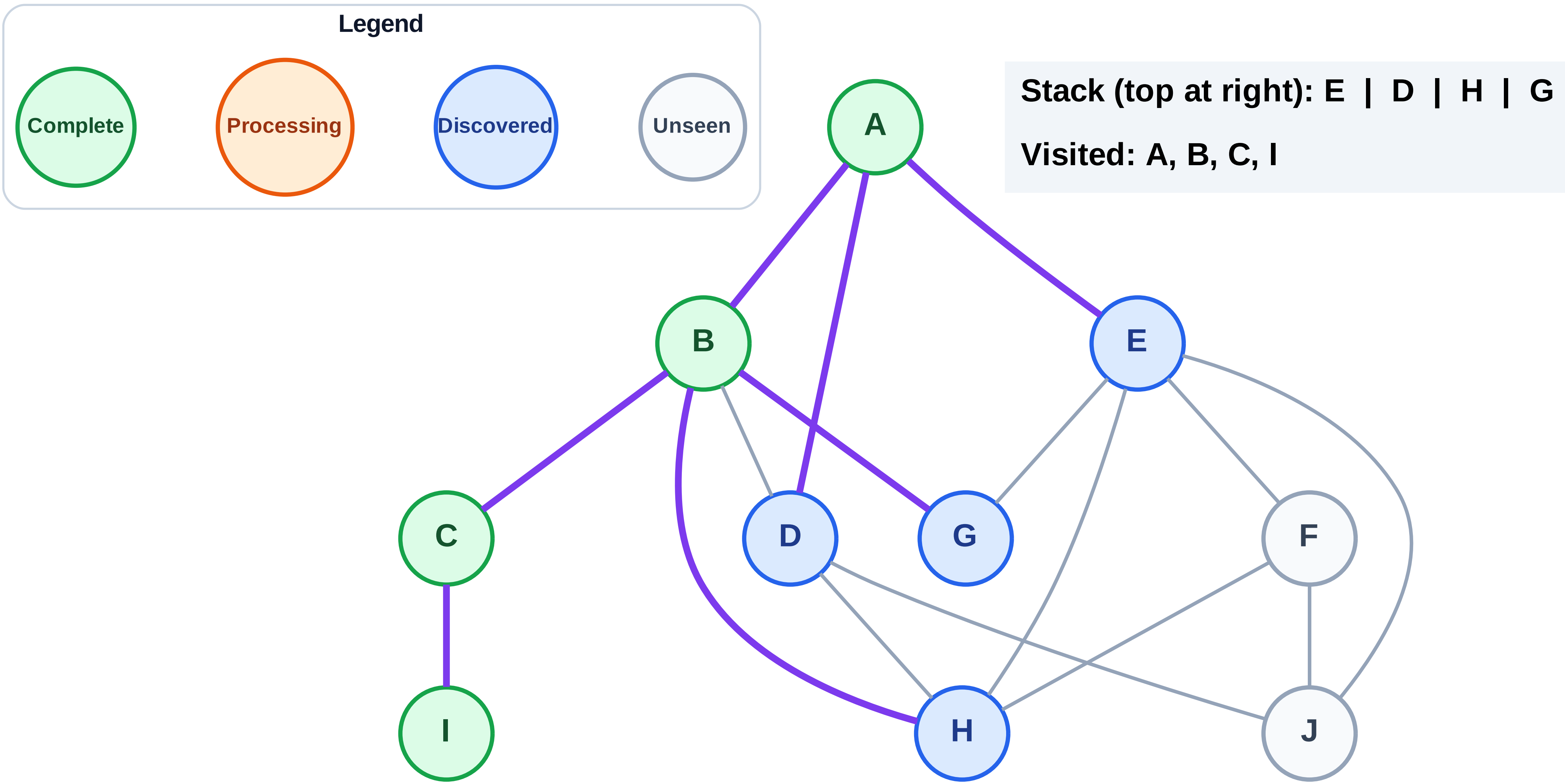


Stack (top at right): E | D | H | G

Visited: A, B, C

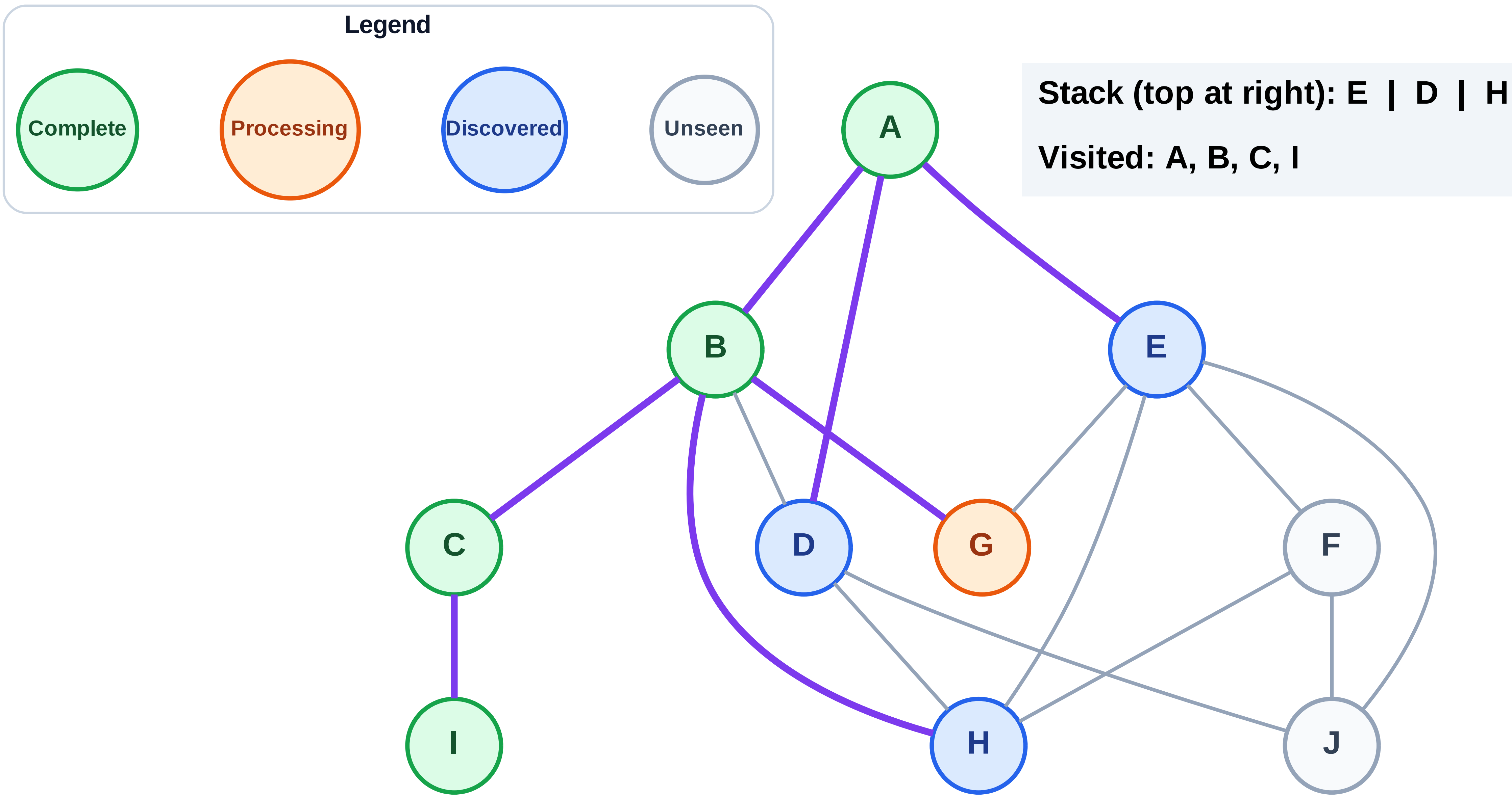
DFS Traversal

Step 9: No new neighbors from I



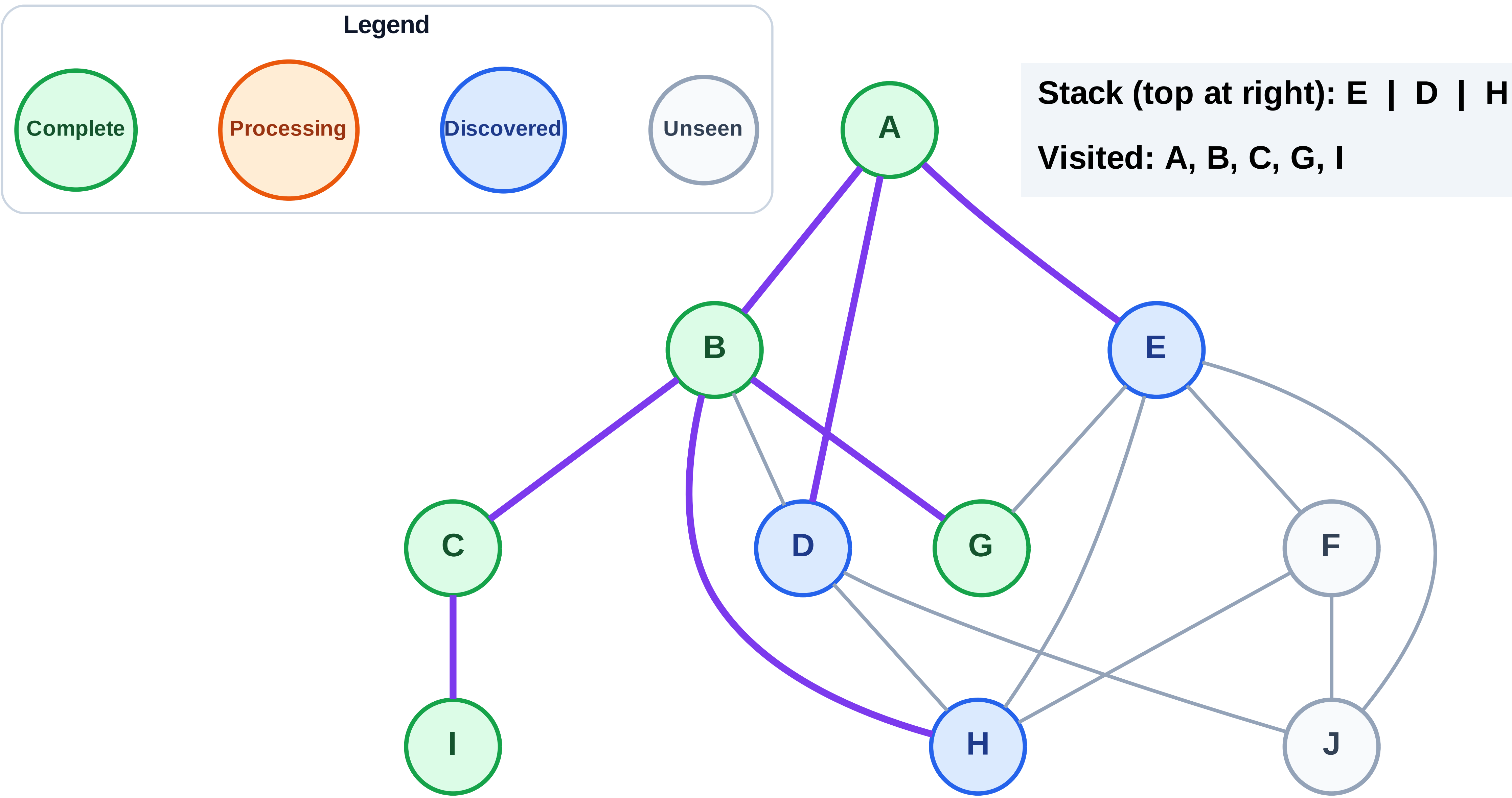
DFS Traversal

Step 10: Process node G



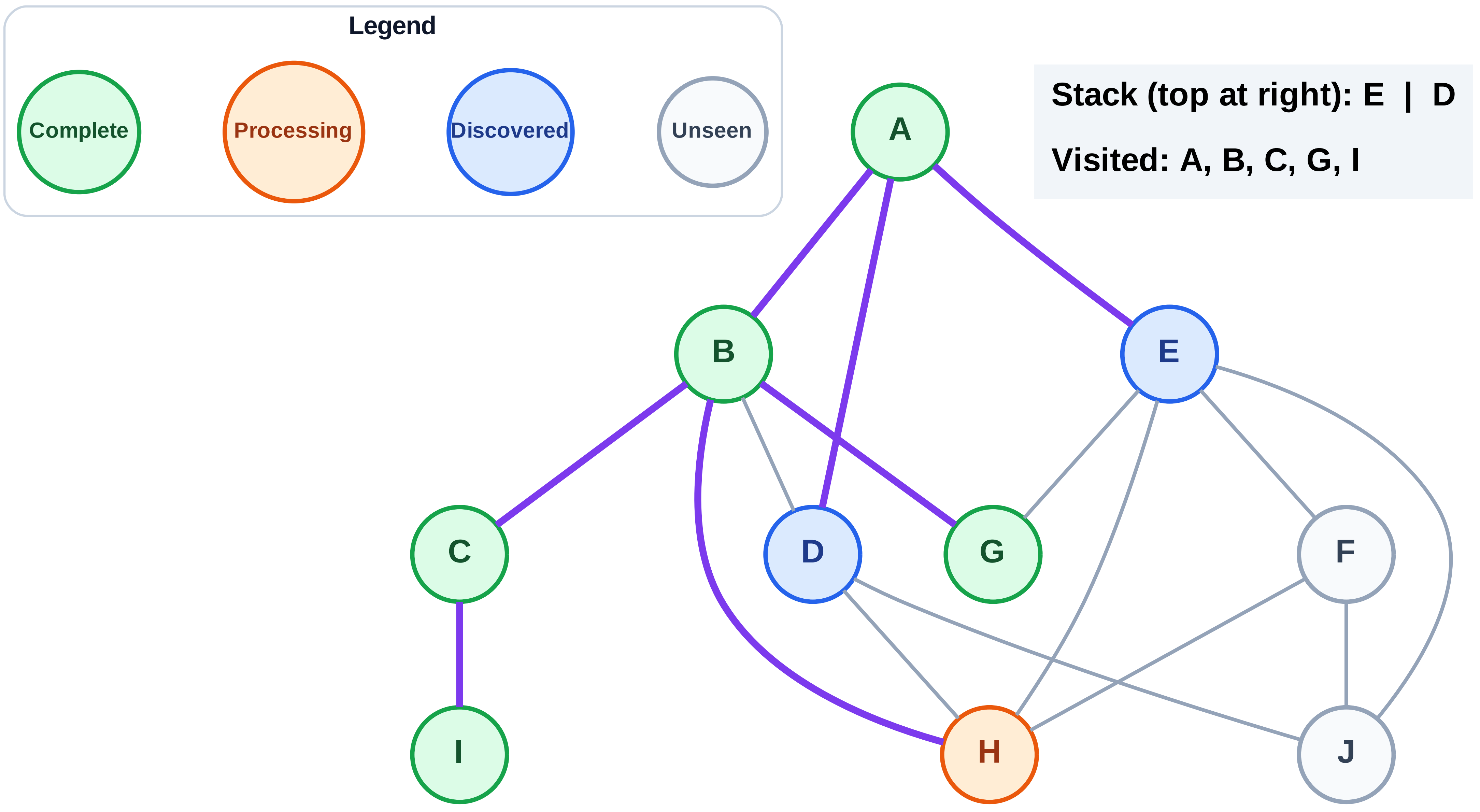
DFS Traversal

Step 11: No new neighbors from G



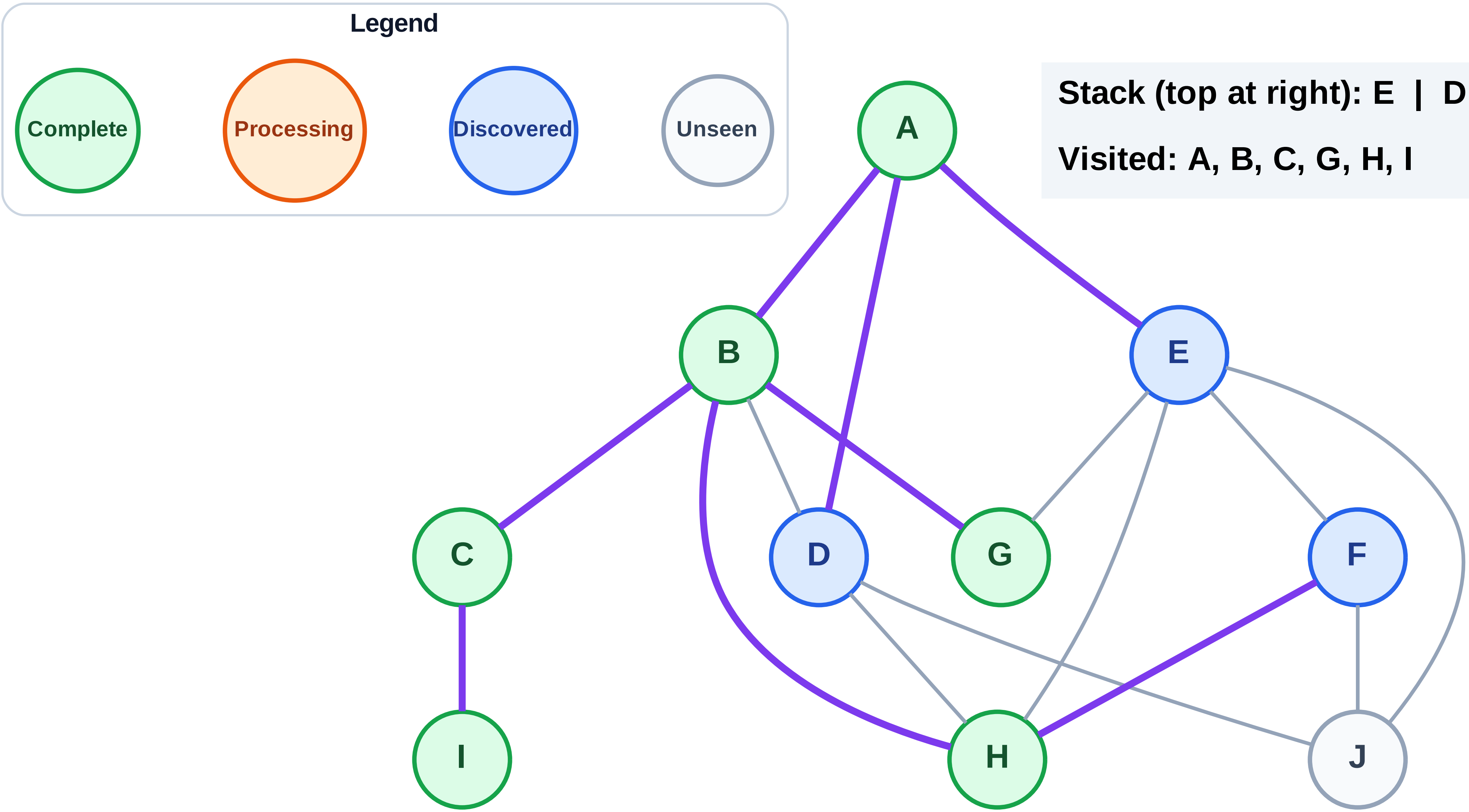
DFS Traversal

Step 12: Process node H



DFS Traversal

Step 13: Discover F from H

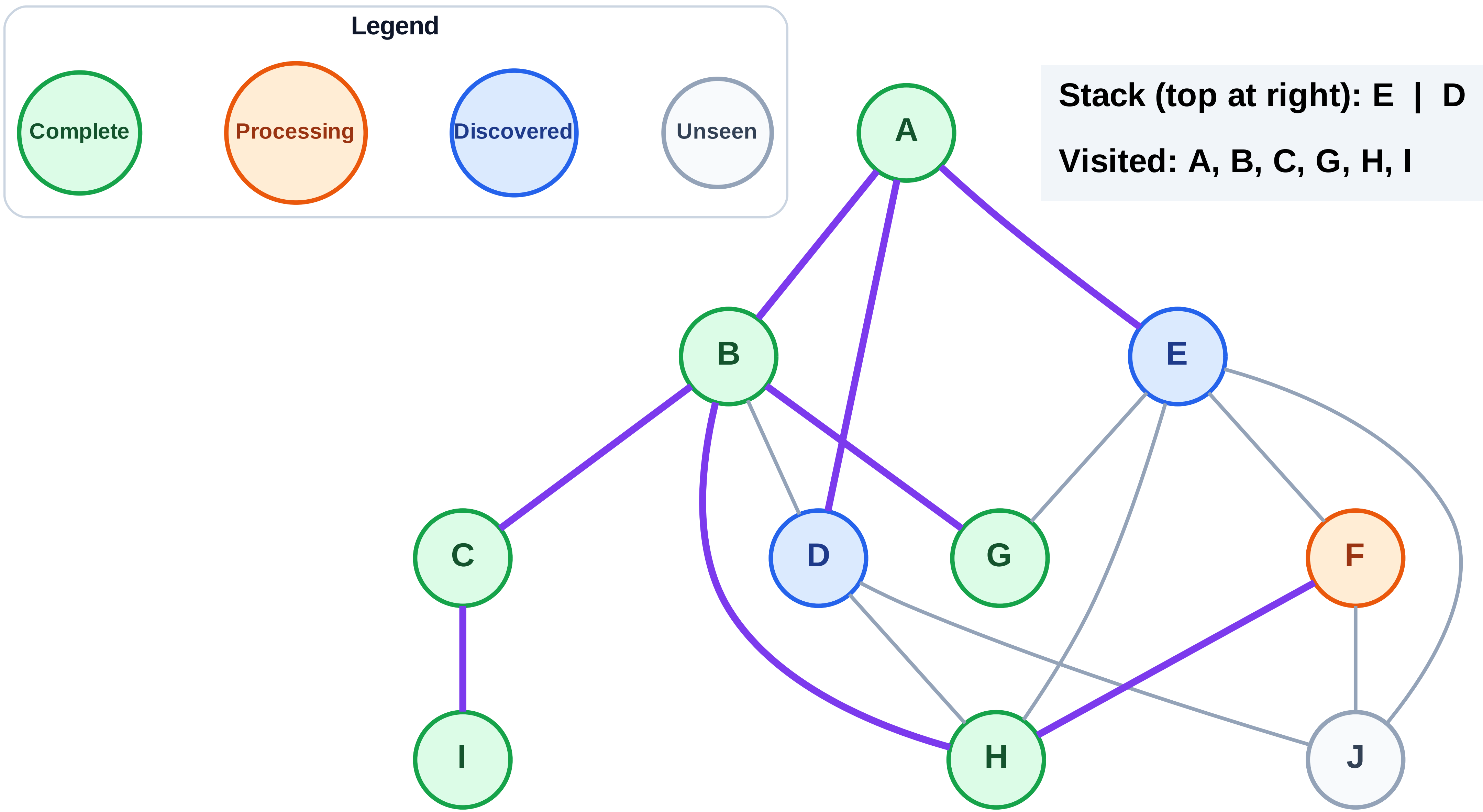


Stack (top at right): E | D | F

Visited: A, B, C, G, H, I

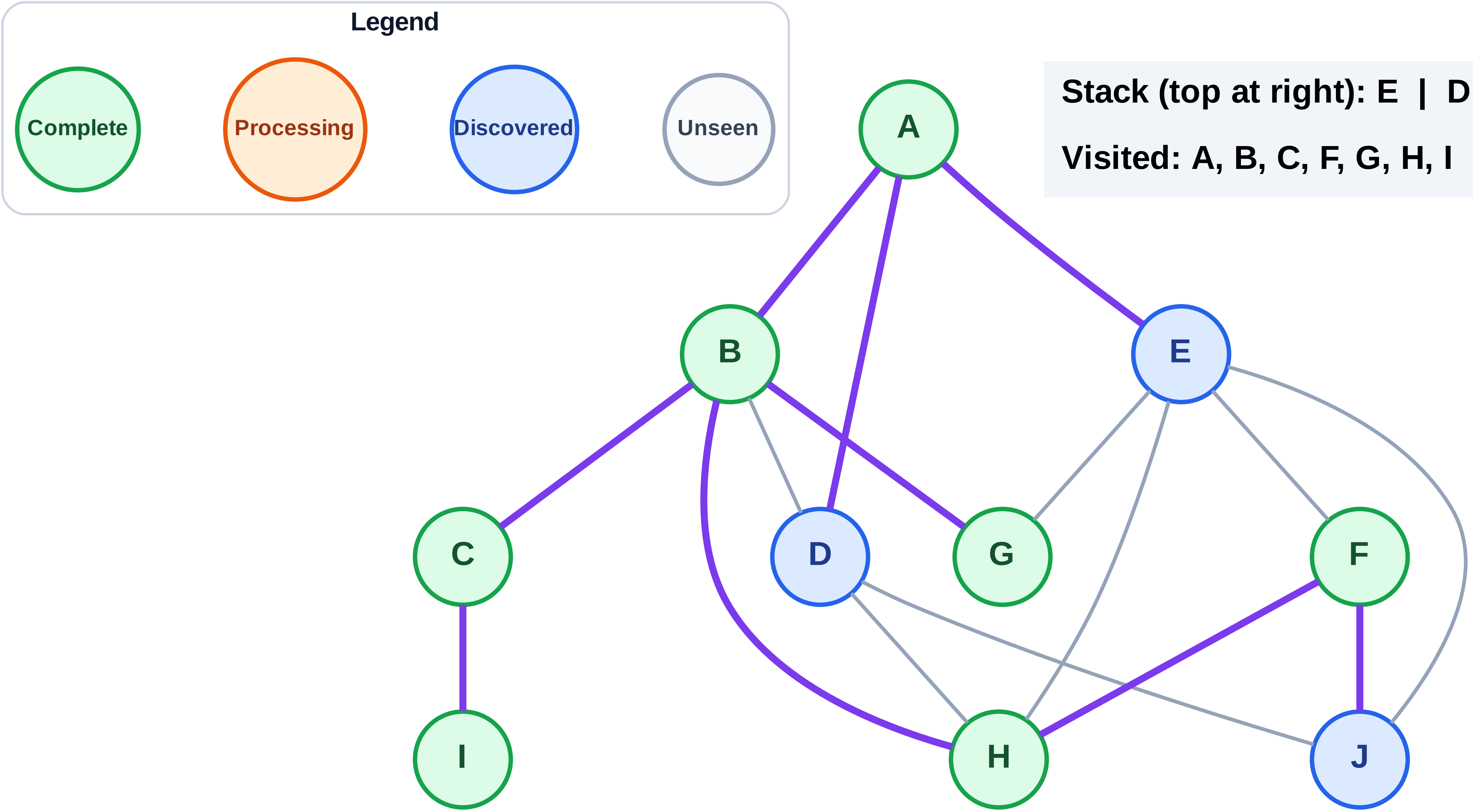
DFS Traversal

Step 14: Process node F



DFS Traversal

Step 15: Discover J from F

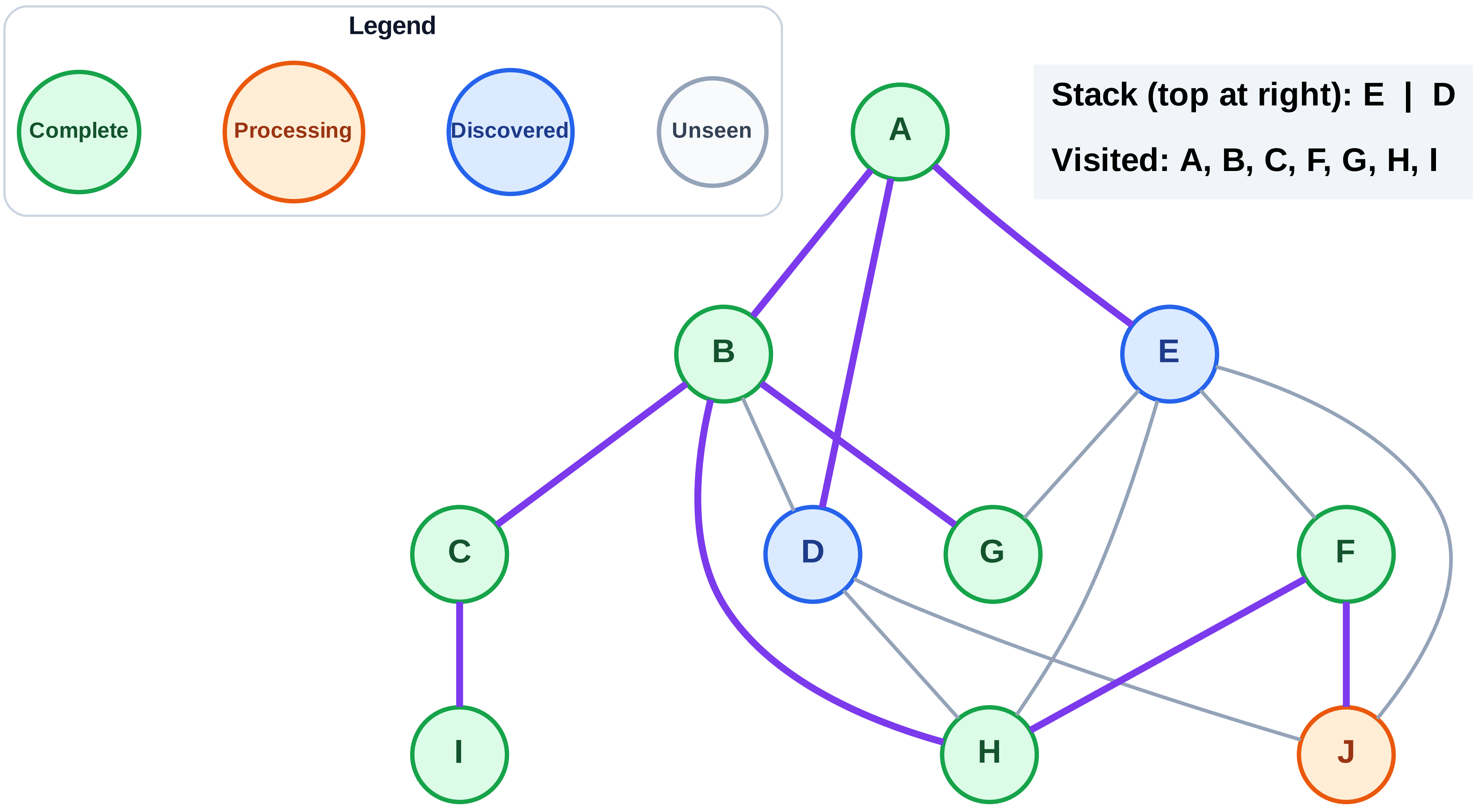


Stack (top at right): E | D | J

Visited: A, B, C, F, G, H, I

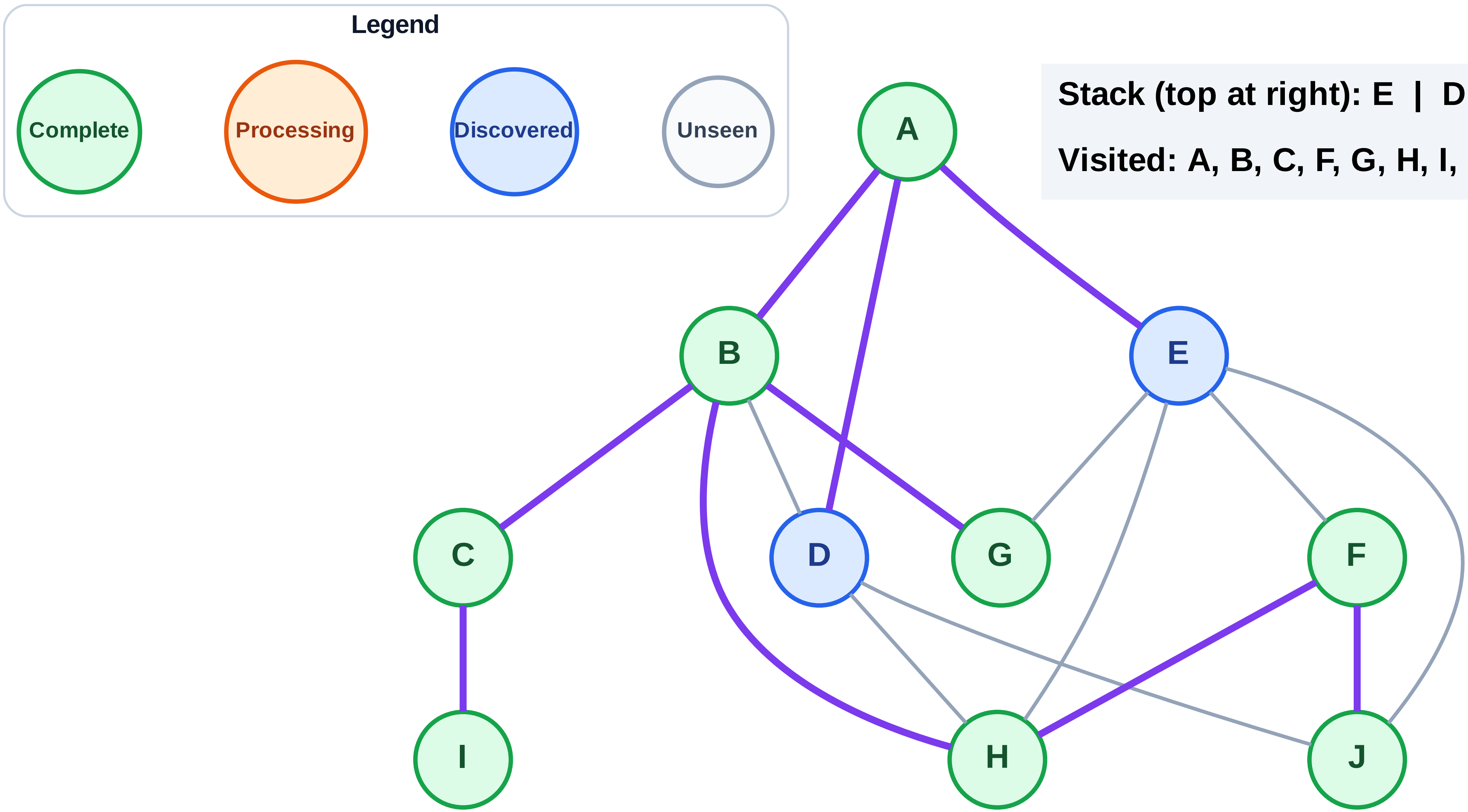
DFS Traversal

Step 16: Process node J



DFS Traversal

Step 17: No new neighbors from J

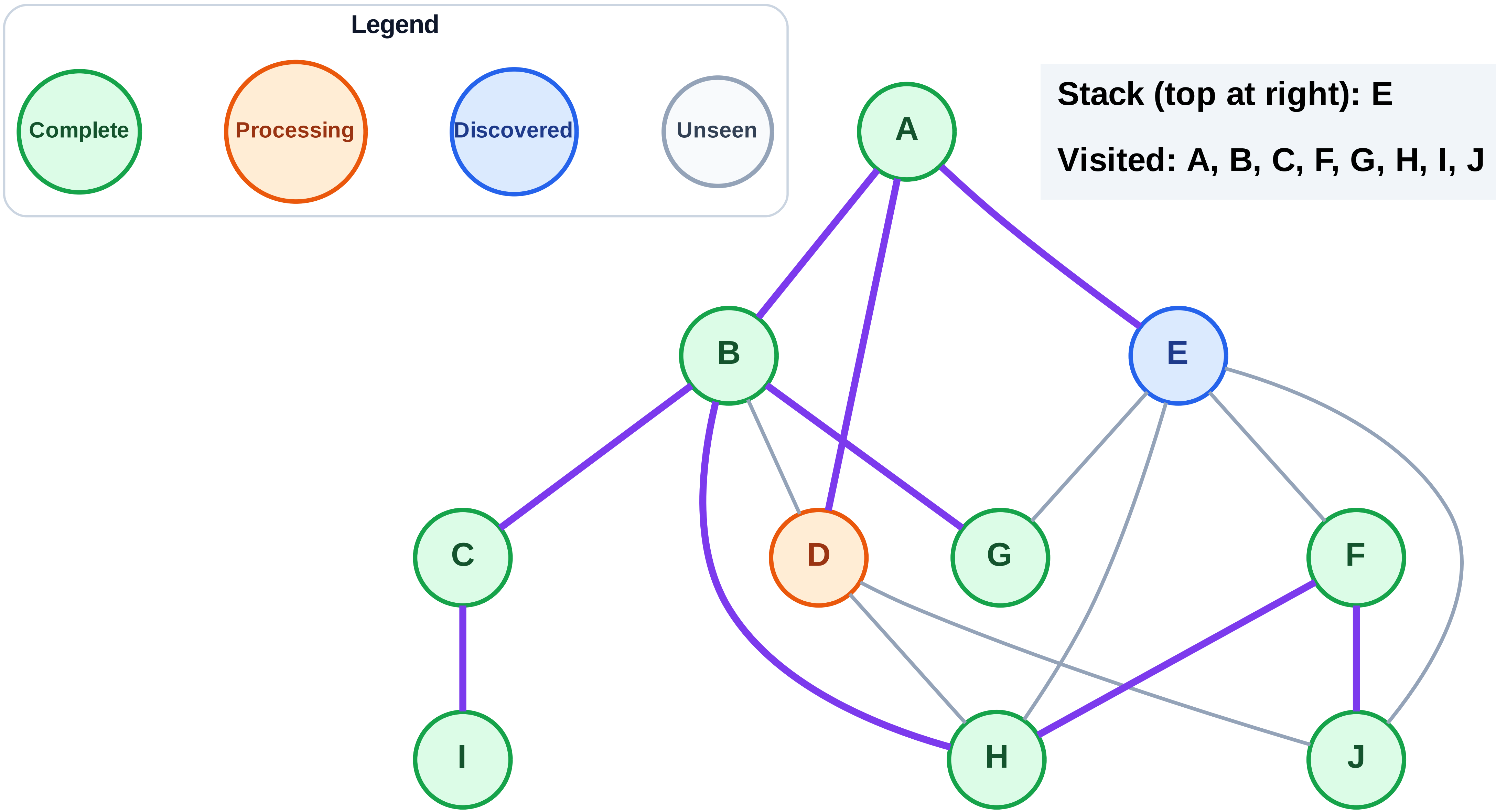


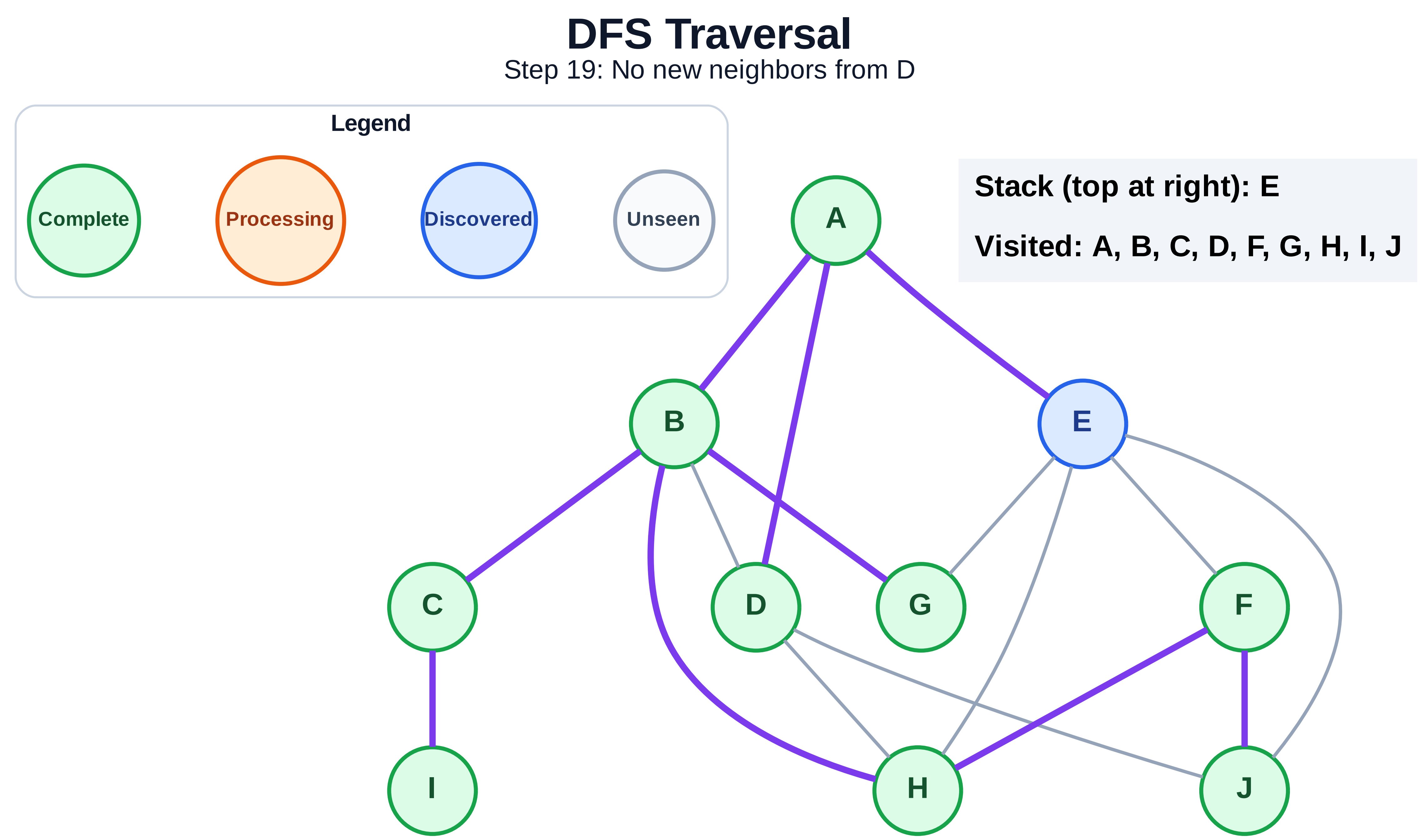
Stack (top at right): E | D

Visited: A, B, C, F, G, H, I, J

DFS Traversal

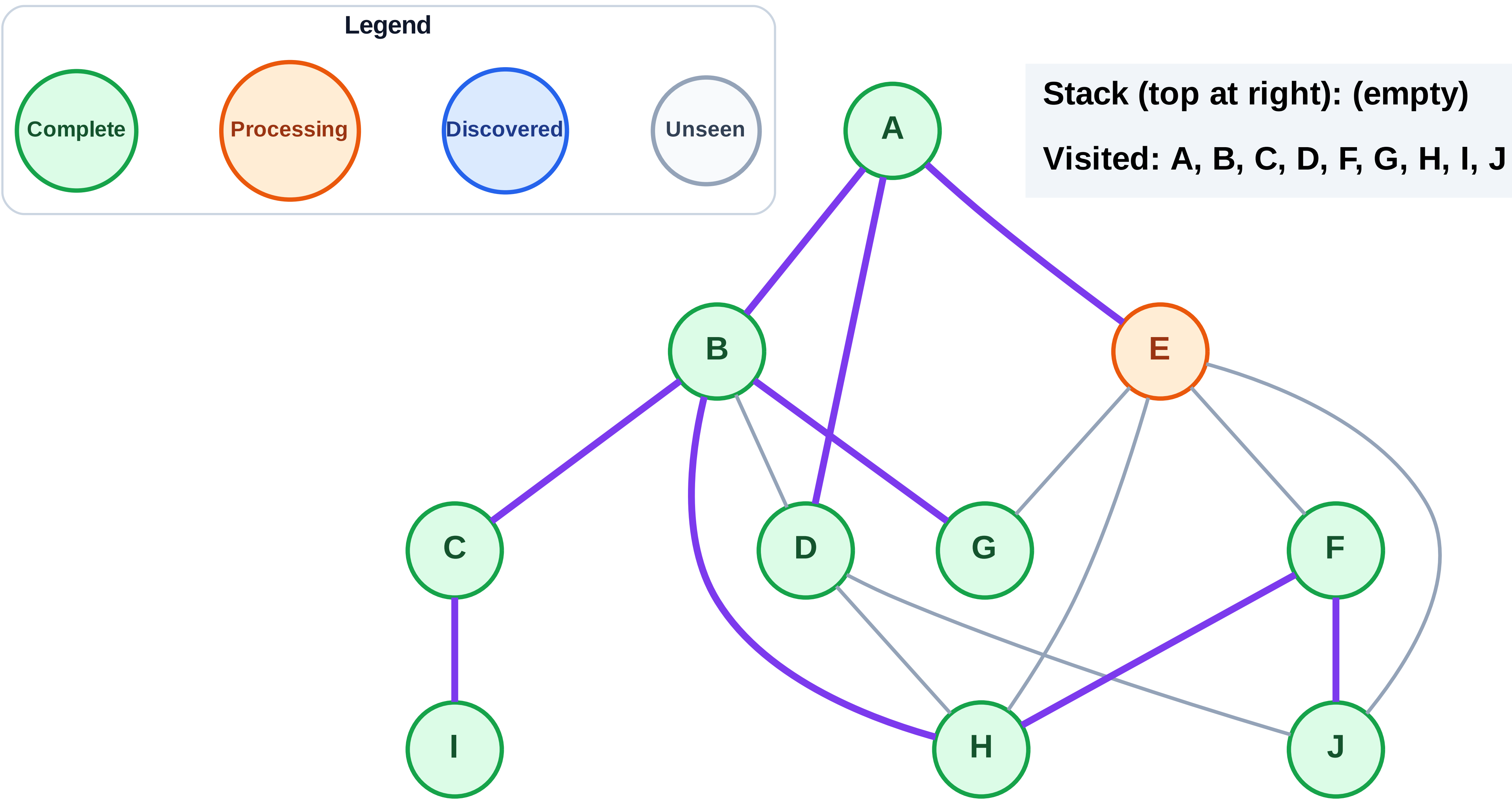
Step 18: Process node D




DFS Traversal
Step 19: No new neighbors from D
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): E
Visited: A, B, C, D, F, G, H, I, J
A
B
E
C
D
G
F
I
H
J

DFS Traversal

Step 20: Process node E



DFS Traversal

Step 21: No new neighbors from E